

Edge Products OpenAPI

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Programming Guide

Intended Audience

This document is intended for device functionality configuration personnel, personnel responsible for alarm reception and reporting integration.

Protocol and Security Specification

HTTP Protocol and Security Specification

Transport Mode	HTTP
Data Format	Requests and responses use JSON format
Character Encoding	UTF-8
Compatible Devices	All interfaces of AIBox and EIA

WebSocket Protocol and Security Specification

Transport Mode	WebSocket
Data Format	Requests and responses use JSON format
Character Encoding	UTF-8
Compatible Devices	AIBox (Configuration + Reporting Interfaces), EIA (Reporting Interface Only)

Parameters sent by third parties when requesting devices via WebSocket:

Parameter	Data Type
method	Request method
uri	Current WebSocket request URI
msgid	Request ID: Used to identify a request-response process

Parameter	Data Type
data	The request data, which is the same as the HTTP request body content

The parameter format for receiving device alarm data via WebSocket by third parties can be found in Chapter 13.

Sample Request Parameters

```
{
  "uri": "/api/v2/search/behavior",
  "method": "POST",
  "msgid": "100001",
  "data": {
    "page": 1,
    "pagesize": 100,
    "channel_id": [1, 2, 3, 4, 5, 6, 7, 8],
    "algo_model": ["FireDetection", "ReflectiveClothesDetectionAlarm", "SmokingAlarm", "TelephoningAlarm", "PlayMobilePhoneDetection", "NoMaskAlarm"],
    "start_time": 1743523200,
    "end_time": 1743609600,
    "object_type": [],
    "plate_no": "",
    "match_type": 999,
    "pic_data_sign": 1,
    "handle_status": [0, 1, 2, 3],
    "vehicle_status": 999
  }
}
```

Explanation of the **response** parameters when third parties request a device via WebSocket:

Parameter	Data Type
code	Response Status Code
msg	Response
msgid	Request ID: Used to identify a request-response process
data	Response data: This part is consistent with the HTTP request body content.

The WebSocket request-response result adds only a “msgid” compared to HTTP; the rest remains the same.

Sample Request Parameters

```
{
  "code": 0,
  "msg": "Succeed",
  "data": {
    "search_id": 1
  },
  "msgid": "100005"
}
```

1 Authentication

To ensure the identity authentication and data security of both parties, a third party must conduct digest authentication when calling this interface.

1.1 Authentication Process

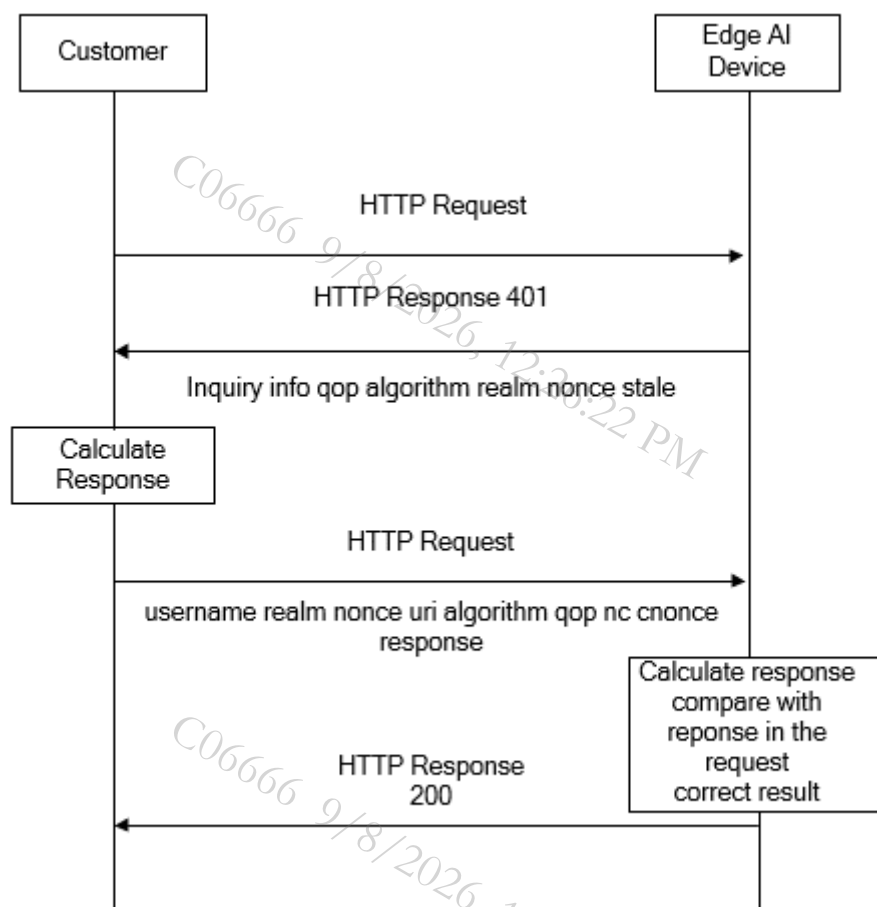


Figure 1.1 Digest Authentication Process

1. When calling any interface without carrying Authorization in the request header, the device returns 401.
The **WWW-Authenticate** field in the response header carries the information needed for authentication: **qop, algorithm, realm, nonce, stale**.
2. Generate digest information based on the returned data fields, place it in the **Authorization** field of the request header, and request the device interface again for a successful request.

1.2 Authentication Information Explanation

Field	Description	Example
username	Username for authentication, assigned by the device.	admin
password	Password for authentication, assigned by the device.	123456
realm	Digest encrypted field, returned by the device during the initial request.	NVRDVR
nonce	Random string, returned by the device during the initial request.	1983339259
uri	URI of the current request.	/api/v2/algo/list
method	Current request method.	POST
algorithm	Digest encryption algorithm, returned by the device during the initial request.	MD5
qop	Quality of protection, returned by the device during the initial request.	auth
nc	Nonce counter, used to prevent replay attacks, generated by the requesting party.	00000002
cnonce	Unique string for additional protection, generated by the requesting party.	1a42d15d190f2f7
response	Digest information for identity verification, generated by the requesting party.	8413c8efa3a4
stale	Indicates whether the nonce from the previous request has expired, returned by the device.	FALSE

Field	Description	Example

1.3 Digest Calculation Process

1. Concatenate **username**, **realm**, **password**, then MD5 hash and encode in hexadecimal, resulting in **r1**.
2. Concatenate **method**, **uri**, then MD5 hash and encode in hexadecimal, resulting in **r2**.
3. Concatenate **r1**, **nonce**, **nc**, **cnonce**, **qop**, **r2**, then MD5 hash and encode in hexadecimal, results in **response**.

1.4 Example

Initial request:

Username provided by the device: admin, password: admin123. Requested URI: /api/v2/algo/list; request method: POST.

Response header info:

```
WWW-Authenticate: Digest  
qop="auth",algorithm="MD5",realm="NVRDVR",nonce="1923473498",stale="FALSE"
```

Response body data:

```
{  
  "code": 3,  
  "msg": "Not Authorized",  
  "status_code": 401  
}
```

Subsequent request:

Request header info:

```
Authorization: Digest username="admin", realm="NVRDVR", nonce="1923473498",  
uri="/api/v2/algo/list", algorithm="MD5", qop=auth, nc=00000002,  
cnonce="1a42d15d190f2f7c93cfe0f749aa4674", response="ebaadf53a48e3156cd5a1485b63658b0"
```

Calculation process:

r1 = hex(md5(username:realm:password))

```
= hex(md5(admin:NVRDVR:admin123.))  
= 98ca67a03cba1d7167bf0096cb957f7d  
r2 = hex(md5(method:uri))  
= hex(md5(POST:/api/v2/algo/list))  
= 305c43426cbcddb71ce670bae226bbc1  
response = hex(md5(r1:nonce:nc:cnonce:qop:r2))  
=hex(md5(98ca67a03cba1d7167bf0096cb957f7d:1923473498:00000002:1  
a42d15d190f2f7c93cfe0f749aa4674:auth:305c43426cbcddb71ce670bae226bb  
c1))  
= ebaadf53a48e3156cd5a1485b63658b0
```

Response body data:

```
{  
  "code": 0,  
  "msg": "Succeed",  
  "status_code": 0,  
  "data": {}  
}
```

2 Person Library Management

2.1 Add Person Library

Basic Info

Path: /api/v2/personlib/add

Method: POST

Interface description:

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks	Other Info
lib_name	string	Yes		Person library name	Max length: 64

Return Data

Name	Type	Required	Default	Remarks
code	integer	Yes		0: Success 400004: Invalid parameter. 400938: Duplicate name. 400921: Failed to add person library
data	object	No		

Name	Type	Required	Default	Remarks
L lib_id	integer	No		Person library ID
msg	string	Yes		

Example

Request:

```
{  
  "lib_name": "test2"  
}
```

Response:

```
{  
  "code": 0,  
  "msg": "Succeed",  
  "status_code": 0,  
  "data": {  
    "lib_id": 5  
  }  
}
```

2.2 Delete Person Library

Basic Info

Path: /api/v2/personlib/delete

Method: POST

Interface description:

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks
lib_id	integer []	Yes		Person library ID

Return Data

Name	Type	Required	Default	Remarks
code	integer	Yes		0: Success 400004: Invalid parameter 400002: Unknown error 400923: Failed to delete person library
msg	string	Yes		

Example

Request:

```
{
  "lib_id": [
    4
  ]
}
```

Response:

```
{
  "code": 0,
  "msg": "Succeed",
  "status_code": 0
}
```

2.3 Modify Person Library Info

Basic Info

Path: /api/v2/personlib/update

Method: POST

Interface description:

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks	Other Info
lib_name	string	Yes		Person library name	Max length: 64
lib_id	integer	Yes		Person library ID	

Return Data

Name	Type	Required	Default	Remarks
code	integer	Yes		0: Success 400004: Invalid parameter 400002: Unknown error 400938: Duplicate name 400925: Failed to modify person library
msg	string	Yes		

Example**Request:**

```
{
  "lib_name": "test4",
  "lib_id": 5
}
```

Response:

```
{
  "code": 0,
  "msg": "Succeed",
  "status_code": 0
}
```

2.4 Get Person Library List

Basic Info

Path: /api/v2/personlib/list**Method:** POST**Interface description:**

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks
lib_name	string	No		Person library name (fuzzy match)

Name	Type	Required	Default	Remarks
page	integer	Yes		Current page
pagesize	integer	Yes		Number of rows per page

Return Data

Name	Type	Required	Default	Remarks	Other Info
code	integer	Yes		0: Success 400004: Invalid parameter 400922: Failed to get person library list	
data	object	Yes			
└ list	object []	Yes			
└─ lib_id	integer	No		Person library ID	
└─ lib_name	string	No		Person library name	Max length: 64
└ total	integer	Yes		Total count	
msg	string	Yes			

Example

Request:

```
{
  "lib_name": "",
  "pagesize": 16,
  "page": 1
}
```

Response:

```
{
```

```
"code": 0,
"msg": "Succeed",
"status_code": 0,
"data": {
  "total": 4,
  "list": [
    {
      "lib_id": 1,
      "lib_name": "placeat"
    },
    {
      "lib_id": 2,
      "lib_name": "test"
    },
    {
      "lib_id": 3,
      "lib_name": "test-2"
    },
    {
      "lib_id": 5,
      "lib_name": "test4"
    }
  ]
}
```

3 Person Management

3.1 Add Person

Basic Info

Path: /api/v2/person/add

Method: POST

Interface description:

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks	Other Info
person_name	string	Yes		Name	Max length: 64
image_base64	string	Yes		Face photo, Base64-encoded	
sex	integer	No		Gender 1: Male 2: Female 99: Unspecified	
email	string	No		Email	Max length: 64
tel	string	No		Contact info	Max length: 32
certificate_type	integer	No		ID type	

Name	Type	Required	Default	Remarks	Other Info
				1: ID card 2: Passport 3: Driver's license 99: Other	
certificate_no	string	No		ID No.	Max length: 64
birth_date	string	No		Birth date	
lib_id	integer	Yes		Person library ID	

Return Data

Name	Type	Required	Default	Remarks
code	integer	Yes		0: Success 400926: Failed to add person 400941: Number of persons exceeds the limit 400004: Invalid parameter 400972: Algorithm error 400930: Failed to operate local image
data	object	No		
└ person_id	integer	No		Person ID
msg	string	Yes		

Example

Request:

```
{  
  "person_name": "bar",  
  "image_base64": "{image_base64}",  
  "sex": 1,  
  "email": "bar@gmail.com",  
}
```

```
"tel": "",
"certificate_type": 1,
"certificate_no": "370109200001010055",
"birth_date": "2000-01-01",
"lib_id": 1
}
```

Response:

```
{
  "code": 0,
  "msg": "Succeed",
  "status_code": 0,
  "data": {
    "person_id": 2007
  }
}
```

3.2 Add Persons in Batches

Basic Info

Path: /api/v2/person/import

Method: POST

Interface description:

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks	Other Info
person_list	object []	Yes			
└ lib_id	integer	Yes		Person library ID	
└ person_name	string	Yes		Name	Max length: 64
└ image_base64	string	Yes		Face photo, Base64-encoded	
└ sex	integer	No		Gender 1: Male 2: Female 99: Unspecified	
└ email	string	No		Email	Max length: 64
└ tel	string	No		Contact info	Max length: 32
└ certificate_type	integer	No		ID type. 1: ID card 2: Passport 3: Driver's license 99: Other	
└ certificate_no	string	No		ID No.	Max length: 64
└ birth_date	string	No		Birth date	

Return Data

Name	Type	Required	Default	Remarks
code	integer	Yes		0: Success 400004: Invalid parameter 403002: Parameter error 400932: Import failed 400002: Unknown error 400940: This part was imported successfully 400941: Number of persons exceeds the limit 400942: No person library is selected
data	object	No		
└ fail_list	string []	No		Person that failed to be imported (name)
msg	string	Yes		

Example

Request:

```
{
  "person_list": [
    {
      "person_name": "bar",
      "image_base64": "{image_base64}",
      "sex": 1,
      "email": "bar@gmail.com",
      "tel": "",
      "certificate_type": 1,
      "certificate_no": "370109200001010055",
      "birth_date": "2000-01-01",
      "lib_id": 1
    },
    {
      "person_name": "buzz",
      "image_base64": "{image_base64}",
      "sex": 1,
      "email": "buzz@gmail.com",
      "tel": "",
      "certificate_type": 1,
    }
  ]
}
```

```
"certificate_no": "",
"birth_date": "2000-01-02",
"lib_id": 1
}
]
```

Response:

```
{
  "code": 0,
  "msg": "Succeed",
  "status_code": 0,
  "data": {
    "fail_list": []
  }
}
```

3.3 Delete Persons in Batches

Basic Info

Path: /api/v2/person/delete**Method:** POST**Interface description:**

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks
person_id_list	integer []	Yes		Face ID

Return Data

Name	Type	Required	Default	Remarks
code	integer	Yes		0: Success 400004: Invalid parameter 400929: Failed to get person list 400927: Failed to delete person
msg	string	Yes		

Example

Request:

```
{
  "person_id_list": [
    2008,
    2006
  ]
}
```

Response:

```
{
  "code": 0,
  "msg": "Succeed",
  "status_code": 0
}
```

3.4 Modify Person Info

Basic Info

Path: /api/v2/person/update

Method: POST

Interface description:

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks	Other Info
person_id	integer	Yes		Person ID	
person_name	string	No		Name	Max length: 64
sex	integer	No		Gender 1: male 2: female 99: unspecified	
email	string	No		Email	Max length: 64
tel	string	No		Contact info	Max length: 32
certificate_type	integer	No		ID type. 1: ID card 2: Passport 3: Driver's license 99: Other	
certificate_no	string	No		ID No.	Max length: 64
image_base64	string	No		Face photo, Base64-encoded	
birth_date	string	No		Birth date	

Return Data

Name	Type	Required	Default	Remarks
code	integer	Yes		0: Success 400004: Invalid parameter 400928: Failed to modify person info 400933: Failed to enable task 400930: Failed to operate local image
msg	string	Yes		

Example

Request:

```
{
  "person_id": 2007,
  "person_name": "bar",
  "image_base64": "{image_base64}",
  "sex": 2,
  "email": "sample@gmail.com",
  "tel": "",
  "certificate_type": 1,
  "certificate_no": "370109200001010055",
  "birth_date": "2000-01-01",
  "lib_id": 1
}
```

Response:

```
{
  "code": 0,
  "msg": "Succeed",
  "status_code": 0
}
```

3.5 Get Person List

Basic Info

Path: /api/v2/person/list

Method: POST

Interface description:

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks	Other Info
lib_id	integer	Yes		Person library ID	
page	integer	Yes		Current page number	
pagesize	integer	Yes		Number of rows per page	
person_name	string	No		Name (supports fuzzy match)	Max length: 64
sex	integer	No		Gender 999: All 1: Male 2: Female 99: Unspecified	
certificate_no	string	No		ID No.	Max length: 64
model_status	integer	No		Modeling status. 999: All, 1: Not started, 2: In progress, 3: Succeeded 4: Failed	

Return Data

Name	Type	Required	Default	Remarks	Other Info
code	string	Yes		0: Success 400004: Invalid parameter 400929: Failed to get person list	
data	object	No			
└ list	object []	No			
└─ person_id	integer	Yes		Person ID	
└─ person_name	string	Yes		Name	Max length: 64
└─ sex	integer	No		Gender 1: Male 2: Female 99: Unspecified	
└─ email	string	No		Email	Max length: 64
└─ tel	string	No		Contact info	Max length: 32
└─ modeling_type	integer	Yes		Modeling status. 0: Not started 2: In progress, 3: Succeeded, 4: Failed	
└─ certificate_type	integer	No		ID type. 1: ID card 2: Passport 3: Driver's	

Name	Type	Required	Default	Remarks	Other Info
				license 99: Other	
└─ certificate_no	string	No		ID No.	Max length: 64
└─ image_path	string	Yes		Image path	Max length: 256
└─ lib_id	integer	Yes		Person library ID	
└─ birth_date	string	No		Birth date	
└─ create_time	integer	No		Creation time	
└─ total	integer	Yes		Total count	
msg	string	Yes			

Example

Request:

```
{
  "lib_id": 1,
  "page": 1,
  "pagesize": 12,
  "sex": 999,
  "person_name": "",
  "certificate_no": "",
  "model_status": 999
}
```

Response:

```
{
  "code": 0,
  "msg": "Succeed",
  "status_code": 0,
  "data": {
    "total": 2,
  }
}
```

```
"list": [  
  {  
    "person_id": 2001,  
    "person_name": "1",  
    "sex": 1,  
    "email": "",  
    "tel": "",  
    "modeling_type": 3,  
    "certificate_type": 1,  
    "certificate_no": "",  
    "image_path": "/api/v2/smart/picture?Type=3&Index=ubs_0_0_0_0_2_1716429305/",  
    "lib_id": 1,  
    "birth_date": "",  
    "create_time": 2737353968  
  },  
  {  
    "person_id": 2002,  
    "person_name": "2",  
    "sex": 1,  
    "email": "",  
    "tel": "",  
    "modeling_type": 3,  
    "certificate_type": 1,  
    "certificate_no": "",  
    "image_path": "/api/v2/smart/picture?Type=3&Index=ubs_0_0_0_1_2_1716429313/",  
    "lib_id": 1,  
    "birth_date": "",  
    "create_time": 2737362084  
  }  
]  
}
```

4 Channel Management

4.1 Get RSA Public Key

Basic Info

Path: /api/v2/rsa/publickey

Method: POST

Interface description:

Request Parameter

Return Data

Name	Type	Required	Default	Remarks	Other Info
code	integer	Yes			
msg	string	Yes			
data	object	Yes			
└ public_key	string	Yes		Public key	

Example

Request:

Response:

```
{
  "code": 0,
  "msg": "Succeed",
  "data": {
    "public_key": "123456789"
  }
}
```

```
}
```

4.2 Get Device Stream Info by Onvif

Basic Info

Path: /api/v2/channel/device/info

Method: POST

Interface description:

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks	Other Info
ip	string	Yes		Camera IP	
port	integer	Yes		Camera port	
username	string	Yes		Camera username	Max length: 64
pwd	string	Yes		Camera password (RSA-encrypted)	Max length: 512

Return Data

Name	Type	Required	Default	Remarks	Other Info
code	integer	Yes			

Name	Type	Required	Default	Remarks	Other Info
data	object	No			
└ manufacturer	string	No		Manufacturer	Max length: 32
└ model	string	No		Model	Max length: 32
└ firmware_version	string	No		Firmware version	Max length: 128
└ serial_number	string	No		Serial number	Max length: 64
└ hardware_id	string	No		Hardware ID	Max length: 32
└ video	object []	Yes		Stream list	
└ └ id	integer	Yes		Stream ID	
└ └ rtsp	string	Yes		RTSP address	Max length: 256
└ └ bitrate	string	No		Bit rate	Max length: 32
└ └ encoding	string	No		Encoding format	Max length: 32
└ └ fps	string	No		Frame rate	Max length: 32
└ └ gop	string	No		Group of picture	Max length: 32
└ └ quality	string	No		Quality	Max length: 32
└ └ ratio	string	No		Resolution	Max length: 32
msg	string	Yes			

Example**Request:**

```
{
  "ip": "217.1.0.143",
  "port": 80,
  "username": "admin",
  "pwd": "gMsL3NeLw=="
}
```

Response:

```
{
  "code": 0,
  "msg": "Succeed",
  "status_code": 0,
  "data": {
    "manufacturer": "HD",
    "model": "A27",
    "firmware_version": "IPC_C6101-B0001P80L01D1809NB",
    "serial_number": "18980000000000V00001",
    "hardware_id": "A27",
    "video": [
      {
        "id": 1,
        "rtsp": "rtsp://217.1.0.143/media/video1",
        "bitrate": "1024",
        "fps": "25",
        "encoding": "H265",
        "gop": "25",
        "quality": "5",
        "ratio": "1920*1080"
      },
      {
        "id": 2,
        "rtsp": "rtsp://217.1.0.143/media/video2",
        "bitrate": "1024",
        "fps": "25",
        "encoding": "H264",
        "gop": "50",
        "quality": "5",

```

```
    "ratio": "720*576"
  },
  {
    "id": 3,
    "rtsp": "rtsp://217.1.0.143/media/video3",
    "bitrate": "128",
    "fps": "25",
    "encoding": "H264",
    "gop": "50",
    "quality": "5",
    "ratio": "352*288"
  }
]
}
```

4.3 Add Channel

Basic Info

Path: /api/v2/channel/add

Method: POST

Interface description:

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks	Other Info
type	integer	Yes		Video protocol 1: onvif 2: rtsp	
transport_type	integer	No		Transport protocol (required for AIBOX, not required for VMS) 1: tcp 2: udp 3: lapi	
channel_name	string	Yes		Channel name	Max length: 64 Min length: 1
rtsp	string	No		RTSP address (required when the video protocol is RTSP).	Max length: 256
ip	string	No		Camera IP address (required when adding camera via Onvif).	
port	integer	No		Camera port (required when adding camera via Onvif).	
username	string	No		Camera username	Max length: 64
pwd	string	No		Camera password (RSA-encrypted)	Max length: 512
video_type	integer	No		Video stream type of Onvif device. 1: main stream 2: custom stream 3: Third stream (only supported by LAPI devices)	
video_id	integer	No		Stream ID: Required when custom stream	

Name	Type	Required	Default	Remarks	Other Info
				is selected. The value is the stream ID returned by the Get Device Stream Info interface.	
custom_code	string	No		Custom code	Max length: 64

Return Data

Name	Type	Required	Default	Remarks	Other Info
code	integer	Yes			
data	object	No			
└ channel_id	integer	Yes		Channel ID	
msg	string	Yes			

Example

Request:

```
{
  "ip": "217.1.0.143",
  "type": 1,
  "port": 80,
  "username": "admin",
  "pwd": "pwdw==",
  "rtsp": "rtsp://217.1.0.143/media/video1",
  "transport_type": 1,
  "channel_name": "Camera",
  "serialnumber": "189802C04B7201V00081",
  "manufacturer": "HD",
  "model": "A27"
}
```

Response:

```
{
  "code": 0,
  "msg": "Succeed",
  "status_code": 0,
  "data": {
    "channel_id": 1
  }
}
```

4.4 Delete Channel

Basic Info

Path: /api/v2/channel/delete

Method: POST

Interface description:

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks
channel_id_list	integer []	Yes		

Return Data

Name	Type	Required	Default	Remarks
code	integer	Yes		0: Success 406303: Parameter error 400304: Failed to delete channel
msg	string	Yes		

Example

Request:

```
{
```

```
  "channel_id_list": [  
    1  
  ]
```

```
}
```

Response:

```
{
```

```
  "code": 0,  
  "msg": "Succeed"
```

```
}
```

4.5 Modify Channel Info

Basic Info

Path: /api/v2/channel/update

Method: POST

Interface description:

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks	Other Info
channel_id	integer	Yes		Channel ID	
transport_type	integer	No		Transport protocol (required for AIBOX, not required for VMS) 1: tcp 2: udp 3: lapi	
channel_name	string	Yes		Channel name	Max length: 64
rtsp	string	No		RTSP address (required when the video protocol is RTSP).	Max length: 256
ip	string	No		Camera IP address (required when adding camera via Onvif).	
port	string	No		Camera port (required when adding camera via Onvif).	
username	string	No		Camera username	Max length: 64
pwd	string	No		Camera password (RSA-encrypted)	Max length: 512
video_type	integer	No		Video stream type of Onvif device. 1: main stream 2: custom stream 3: Third stream (Only	

Name	Type	Required	Default	Remarks	Other Info
				supported by LAPI devices)	
video_id	integer	No		Stream ID: Required when custom stream is selected. The value is the stream ID returned by the Get Device Stream Info interface.	
custom_code	string	No		Custom code. Length: 64	Max length: 64

Return Data

Name	Type	Required	Default	Remarks
code	integer	Yes		0: Success 406305: Parameter error 400319: Channel name cannot be empty 400318: Channel name cannot include spaces 400317: Channel name cannot contain more than 10 characters 400306: Failed to edit channel 400315: Duplicate channel name
msg	string	Yes		

Example

Request:

```
{  
  "channel_id": 1,  
  "channel_name": "Camera1",  
  "status": 1,  
  "ip": "217.1.0.143",  
  "port": 80,  
  "type": 1,  
}
```

```

    "username": "admin",
    "off_reason": 1,
    "manufacturer": "HD",
    "model": "A27",
    "serial_number": "189802C04B7201V00081",
    "algo_model": "",
    "Nums": 1,
    "pwd": "pwdQ==",
    "serialnumber": "189802C04B7201V00081",
    "rtsp": "rtsp://217.1.0.143/media/video1"
  }

```

Response:

```

{
  "code": 0,
  "msg": "Succeed",
  "status_code": 0
}

```

4.6 Get Channel List

Basic Info

Path: /api/v2/channel/list

Method: POST

Interface description:

Only the EIA device type supports searching channel list based on conditions.

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks	Other Info
channel_name	string	No		Channel name (supports fuzzy match)	Max length: 64
page	integer	Yes		Current page	
pagesize	integer	Yes		Number of rows per page	

Return Data

Name	Type	Required	Default	Remarks	Other Info
code	string	Yes		0: success 400300: failed to get channel list	
data	object	Yes			
└ total	integer	Yes		Total count	
└ channel_list	object []	No		Channel list	
└─ channel_id	integer	Yes		Channel ID	

Name	Type	Required	Default	Remarks	Other Info
└─ custom_code	string	No		Custom code	Max length: 64
└─ channel_name	string	Yes		Channel name	Max length: 64
└─ type	integer	Yes		Video stream protocol 1: onvif 2: rtsp	
└─ transport_type	integer	No		Transport protocol (required for AIBOX, not required for VMS) 1: tcp 2: udp	
└─ ip	string	No		Channel IP	
└─ port	number	No		Channel port	
└─ serial_number	string	No		Camera's serial number	Max length: 64
└─ model	string	No		Model	Max length: 32
└─ manufacturer	string	No		Manufacturer	Max length: 32
└─ status	integer	Yes		1: Online 2: Offline	
└─ rtsp	string	No		RTSP address	

Name	Type	Required	Default	Remarks	Other Info
flv	string	No		Flv address (returned by VMS)	
msg	string	Yes			

Example

Request:

```
{
  "page": 1,
  "pagesize": 10,
  "channel_name": ""
}
```

Response:

```
{
  "code": 0,
  "msg": "Succeed",
  "status_code": 0,
  "data": {
    "total": 10,
    "channel_list": [
      {
        "channel_id": 1,
        "channel_name": "Camera1",
        "status": 1,
        "ip": "217.1.0.143",
        "port": 80,
        "type": 1,

```

```
{
  "username": "admin",
  "off_reason": 1,
  "manufacturer": "HD",
  "model": "A27",
  "serial_number": "189802C000000V00001"
}
```

4.7 Add Channels in Batches

Basic Info

Path: /api/v2/channel/import

Method: POST

Interface description:

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks	Other Info
channel_list	object []	Yes		Channel list	
└ type	integer	Yes		Video stream protocol 1: onvif 2: rtsp	

Name	Type	Required	Default	Remarks	Other Info
└ transport_type	integer	No		Transport protocol (required for AIBOX, not required for VMS) 1: tcp 2: udp	
└ channel_name	string	Yes		Channel name	Max length: 64
└ rtsp	string	No		Video stream RTSP address	Max length: 256
└ ip	string	No		Camera IP address (required when adding camera via Onvif).	
└ port	string	No		Camera port (required when adding camera via Onvif).	
└ username	string	No		Camera username	Max length: 64
└ pwd	string	No		Camera password (RSA-encrypted)	Max length: 512
└ video_type	integer	No		Video stream type of Onvif device. 1: main stream 2: custom stream	
└ video_id	integer	No		Stream ID: Required when custom stream is selected. The value is the stream ID returned by the Get Device	

Name	Type	Required	Default	Remarks	Other Info
				Stream Info interface.	
└ custom_code	string	Yes		Custom code	Max length: 64

Return Data

Name	Type	Required	Default	Remarks
code	integer	Yes		0: Success 400307: Failed to import template for adding channels in batches 400307: Error occurred when importing template for adding channels in batches 400316: Number of channels exceeds the limit
data	object	No		
└ fail_list	string []	No		Channel that failed to be imported (name)
msg	string	Yes		

Example

Request:

```
{
  "channel_list": [
    {
      "ip": "217.1.0.143",
      "type": 1,
      "port": 80,
      "username": "admin",
      "pwd": "pwdw==",
      "rtsp": "rtsp://217.1.0.143/media/video1",
      "transport_type": 1,
    }
  ]
}
```

```
"channel_name": "Camera",
"serialnumber": "189802C04B7201V00081",
"manufacturer": "HD",
"model": "A27"
},
{
  "ip": "217.1.0.144",
  "type": 1,
  "port": 80,
  "username": "admin",
  "pwd": "pwdw==",
  "rtsp": "rtsp://217.1.0.144/media/video1",
  "transport_type": 1,
  "channel_name": "Camera",
  "serialnumber": "189802C04B7201V00082",
  "manufacturer": "HD",
  "model": "A27"
}
]
}
```

Response:

```
{
  "code": 0,
  "msg": "Succeed",
  "data": {
    "fail_list": []
  }
}
```

5 Channel Arming Time

5.1 Add/Modify Arming Time

Basic Info

Path: /api/v2/control/time/update

Method: POST

Interface description:

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks
channel_id	integer	Yes		Channel ID
algo_model	string	Yes		Algorithm model
control_time	object	Yes		Arming time
└ monday	object []	Yes		
└─ start	string	No		Example: 00:00
└─ end	string	No		Example: 24:00
└ tuesday	object []	Yes		
└─ start	string	No		
└─ end	string	No		
└ wednesday	object []	Yes		
└─ start	string	No		

Name	Type	Required	Default	Remarks
└─ end	string	No		
└─ thursday	object []	Yes		
└─ start	string	No		
└─ end	string	No		
└─ friday	object []	Yes		
└─ start	string	No		
└─ end	string	No		
└─ saturday	object []	Yes		
└─ start	string	No		
└─ end	string	No		
└─ sunday	object []	Yes		
└─ start	string	No		
└─ end	string	No		

Return Data

Name	Type	Required	Default	Remarks
code	integer	Yes		
msg	string	Yes		

Example

Request:

```
{
  "channel_id": 1,
  "algo_model": "FallOverAlarm",
  "control_time": {
    "monday": [
```

```
{
  "start": "00:00",
  "end": "04:00"
},
{
  "start": "12:30",
  "end": "24:00"
}
],
"tuesday": [
  {
    "start": "00:00",
    "end": "05:00"
  },
  {
    "start": "15:00",
    "end": "24:00"
  }
],
"wednesday": [
  {
    "start": "00:00",
    "end": "05:00"
  },
  {
    "start": "15:30",
    "end": "24:00"
  }
],
"thursday": [
  {
    "start": "00:00",
    "end": "04:00"
  },
  {
    "start": "12:30",
    "end": "24:00"
  }
]
```

```
],
  "friday": [
    {
      "start": "00:00",
      "end": "04:00"
    },
    {
      "start": "12:30",
      "end": "24:00"
    }
  ],
  "saturday": [
    {
      "start": "00:00",
      "end": "10:00"
    },
    {
      "start": "14:30",
      "end": "24:00"
    }
  ],
  "sunday": [
    {
      "start": "00:00",
      "end": "10:00"
    },
    {
      "start": "14:30",
      "end": "24:00"
    }
  ]
]
}
```

Response:

```
{
  "code": 0,
```

```
"msg": "Succeed",  
"status_code": 0  
}
```

5.2 Get Arming Time

Basic Info

Path: /api/v2/control/time/get

Method: POST

Interface description:

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks
channel_id	integer	Yes		Channel ID
algo_model	string	Yes		Algorithm model

Return Data

Name	Type	Required	Default	Remarks
code	integer	Yes		
data	object	No		

Name	Type	Required	Default	Remarks
└─ monday	object []	Yes		
└─┬─ start	string	No		
└─┬─ end	string	No		
└─ tuesday	object []	Yes		
└─┬─ start	string	No		
└─┬─ end	string	No		
└─ wednesday	object []	Yes		
└─┬─ start	string	No		
└─┬─ end	string	No		
└─ thursday	object []	Yes		
└─┬─ start	string	No		
└─┬─ end	string	No		
└─ friday	object []	Yes		
└─┬─ start	string	No		
└─┬─ end	string	No		
└─ saturday	object []	Yes		
└─┬─ start	string	No		
└─┬─ end	string	No		
└─ sunday	object []	Yes		
└─┬─ start	string	No		
└─┬─ end	string	No		
msg	string	Yes		

Example

Request:

```

{
  "channel_id": 1,

```

```
"algo_model": "FallOverAlarm"
}
```

Response:

```
{
  "code": 0,
  "msg": "Succeed",
  "status_code": 0,
  "data": {
    "monday": [
      {
        "start": "00:00",
        "end": "04:00"
      },
      {
        "start": "12:30",
        "end": "24:00"
      }
    ],
    "tuesday": [
      {
        "start": "00:00",
        "end": "05:00"
      },
      {
        "start": "15:00",
        "end": "24:00"
      }
    ],
    "wednesday": [
      {
        "start": "00:00",
        "end": "05:00"
      },
      {
        "start": "15:30",
        "end": "24:00"
      }
    ]
  }
}
```

```
    }  
  ],  
  "thursday": [  
    {  
      "start": "00:00",  
      "end": "04:00"  
    },  
    {  
      "start": "12:30",  
      "end": "24:00"  
    }  
  ],  
  "friday": [  
    {  
      "start": "00:00",  
      "end": "04:00"  
    },  
    {  
      "start": "12:30",  
      "end": "24:00"  
    }  
  ],  
  "saturday": [  
    {  
      "start": "00:00",  
      "end": "10:00"  
    },  
    {  
      "start": "14:30",  
      "end": "24:00"  
    }  
  ],  
  "sunday": [  
    {  
      "start": "00:00",  
      "end": "10:00"  
    },  
    {
```

```
"start": "14:30",  
"end": "24:00"  
}  
]  
}  
}
```

C06666 9/8/2026, 12:26:22 PM

C06666 9/8/2026, 12:26:22 PM

C06666 9/8/2026, 12:26:22 PM

C06666 9/8/2026, 12:26:22 PM

6 Intelligent Analysis Configuration

6.1 Add (Modify) Intelligent Analysis Task

Basic Info

Path: /api/v2/smart/update

Method: POST

Interface description:

Fuel unloading algorithm service name

1. TankerParking Fuel Tanker Parking
2. SafetyParking Wheel Chock Placement
3. SafetyOperation Safety Operations
4. SafetyEquipment Safety Device Placement
5. TubingConnection Fuel Hose Connection
6. PersonOnDuty Personnel On Duty
7. TubingDisconnection Post-Unloading Confirmation

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks
smart_list	object []	Yes		Intelligent analysis task list
└ algo_model	string	No		Algorithm model. Max length: 64
└ time_threshold	integer	No		Time threshold (second)
└ person_num_limit	integer	No		Person number limit
└ person_float_limit	integer	No		Fluctuating number of people
└ back_time_threshold	integer	No		Background refresh interval (second)
└ line_dpc_enable	integer	No		Whether to enable tripwire people counting reset (VMS does not support this parameter) 1: enabled 0: disabled
└ line_dpc_time	integer	No		Time when tripwire people counting results will be reset: Integer, in the range of 0-23 (VMS does not support this parameter).
└ detour_area	integer	No		Whether is a detour area. 0: No 1: Yes

Name	Type	Required	Default	Remarks
└ object_type	integer []	No		Type of intrusion detection target. 1: Pedestrian 2: Non-motor vehicle 3: Motor vehicle Multiple options are allowed. 5 Bear, 6 Wild Boar, 7 Tiger, 8 Horse, 9 Cattle, 10 Pig, 11 Sheep, 12 Cat, 13 Dog, 14 Leopard, 15 Wolf, 16 Deer, 17 Kangaroos, 99 Others
└ max_person_object_height	integer	No		Maximum pedestrian height. Minimum: 0 Maximum: Less than 1080.
└ max_person_object_width	integer	No		Maximum pedestrian width. Minimum: 0 Maximum: Less than 1920.
└ min_person_object_height	integer	No		Minimum pedestrian height. Minimum: 0 Maximum: Less than 1080.
└ min_person_object_width	integer	No		Minimum pedestrian width. Minimum: 0 Maximum: Less than 1920.

Name	Type	Required	Default	Remarks
└ max_vehicle_object_height	integer	No		Maximum motor vehicle height. Minimum: 0 Maximum: Less than 1080
└ max_vehicle_object_width	integer	No		Maximum motor vehicle width. Minimum: 0 Maximum: Less than 1920
└ min_vehicle_object_height	integer	No		Minimum motor vehicle height. Minimum: 0 Maximum: Less than 1080
└ min_vehicle_object_width	integer	No		Minimum motor vehicle width. Minimum: 0 Maximum: Less than 1920
└ max_nonVehicle_object_height	integer	No		Maximum non-motor vehicle height. Minimum: 0 Maximum: Less than 1080
└ max_nonVehicle_object_width	integer	No		Maximum non-motor vehicle width. Minimum: 0 Maximum: Less than 1920
└ min_nonVehicle_object_height	integer	No		Minimum non-motor vehicle height. Minimum: 0 Maximum: Less than 1080
└ min_nonVehicle_object_width	integer	No		Minimum non-motor vehicle width

Name	Type	Required	Default	Remarks
				Minimum: 0 Maximum: Less than 1920
└ standard_line	integer	No		Distance measurement line (meter) Used by speeding vehicle analysis algorithms
└ speed_info	object	No		Speeding parameter
└─ default_speed_limit	integer	No		Default speed limit (km/h)
└─ speed_limit_fluctuate	integer	No		Actual speed limit (percentage)
└─ speed_correction_infos	object []	No		Speed correction, three groups
└─ speed_correction	integer	No		Correction speed ratio
└─ speed_max	integer	No		Maximum correction speed
└─ speed_min	integer	No		Minimum correction speed
└─ congestion_info	object	No		Congestion parameter
└─ congestion_time	integer	No		Congestion time
└─ congestion_num	integer	No		Number of congested vehicles
└ enable_work_clothes_alarm	integer	No		Illegal fueling Alarm mode 0: All people

Name	Type	Required	Default	Remarks
				1: Non-staff members
└─ fuel_unloading_info	object	No		Fuel unloading parameter
└─ person_on_duty_overtime	integer			Time when the person leaves the post.
└─ stabilize_time	integer			Fuel stabilization time
└─ incident_model	string			Detection task. See fuel unloading algorithm service name.
└─ park_time	integer	No		Parking time
└─ water_color	string []	No		Abnormal water color, multiple selections allowed White Black Red Yellow Blue Green
└─ high_water	integer	No		Highest water level
└─ low_water	integer	No		Lowest water level
└─ alarmed_coal_quantity	integer	No		Proportion of the belt width taken up by coal blocks (in ten-thousandths percentage)

Name	Type	Required	Default	Remarks
└ coal_option	string []	No		Coal quantity alarm, multiple selections allowed. Empty: No coal Few: Few coal Normal: Normal quantity Full: Overflow
└ flush_time	integer	No		Background refresh interval
└ dog_info	object	No		Dog detection parameters
└ dog_leash_length	integer	No		Leash length
└ ship_option	string []	No		Ship type, multiple selections allowed. All: All SandShip: Sandship PassengerShip: Passenger ship
└ ship_type	integer []	No		Ship type, multiple selections allowed. Supported by since version 2.3.6 99: Other ships 1: Sand barge detection 2: Cargo ship detection 3: Passenger ship detection

Name	Type	Required	Default	Remarks
└ rise_detect_mode	integer	No		Getting up detection mode 1: Leave 0: Enter
└ report_interval	integer	No		Traffic event reporting interval
└ feature_mode	integer	No		Whether uses the default base library images 1: Yes 0: No
└ fall_mode	integer	No		Fall detection scene type 0: General scene 1: Escalator scene
└ autotimer_info	object	No		Long queue detection parameter
└ auto_hour	integer	No		Hour
└ auto_minute	integer	No		Minute
└ auto_seconds	integer	No		Second
└ auto_direction	integer	No		Enable/disable automatic direction recognition
└ regulation	string	No		Rule 1: $\leq$ Rule 2: $\geq$ Rule 3: $\geq$ and $\leq$ Rule 4: $\geq$ or $\leq$
└ meter_reading_value1	number	No		Value 1
└ meter_reading_value2	number	No		Value 2
└ alarm_condition	integer	No		Switch position:

Name	Type	Required	Default	Remarks
				1: Open 2: Closed
└ algo_model_mode	integer	No		Climbing detection (1: Head-Shoulder Mode, 2: Full-Body Mode) Fight detection (3: Indoor, 4: Outdoor)
└ identity_recognition_param	object[]	No		Identity recognition parameters
└ condition_type	integer	Yes		Condition type: 1: Match All Conditions 2: Match any condition
└ condition_details	object	Yes		Condition
└ regulation	integer	Yes		Rule: Rule 1: $\leq$ Rule 2: $\geq$ Rule 3: $\geq$ and $\leq$ Rule 4: $\geq$ or $\leq$
└ meter_reading_value1	integer	Yes		Value 1
└ meter_reading_value2	integer	Yes		Value 2
└ lib_id	integer	Yes		Work clothes library ID
└ workclothes_lib_list	integer[]	No		Work clothes library ID
└ graphs	object[]	No		Area info (Note: the range of all the coordinates is 0~10000)

Name	Type	Required	Default	Remarks
└─ graph_type	string	No		Area type. Area: polygon Tripwire: line Mixed: blend
└─ polygon	object	No		Area coordinates. When the area type is polygon or blend, this field represents the coordinates of the drawn area.
└─ detour_area	integer	No		Whether is a detour area. 0: No 1: Yes
└─ fuel_task_type	string			Fuel unloading detection type
└─ graph_usage	string	No		ROI: Analyzing area NotROI: Shield area
└─ point_x	string	No		Horizontal coordinates. When the area type is polygon or blend, it is required to draw 3-6 points; format: x1, x2, x3, x4, x5, x6
└─ point_y	string	No		Vertical coordinates. When the area type is polygon or blend, it is required to draw 3-6 points; format: y1, y2, y3, y4, y5, y6

Name	Type	Required	Default	Remarks
 line	object	No		Tripwire coordinates. When the area type is line or blend, this field represents the coordinates of the tripwire.
 point_x	string	No		Horizontal coordinates. When the area type is line or blend, it is required to draw 2 points; format: x1,x2
 point_y	string	No		Vertical coordinates. When the area type is line or blend, it is required to draw 2 points; format: y1,y2
 scalevalue	object	No		Specified area detected by the pressure gauge
 point_x	string	No		Horizontal coordinates. Requires at least 2 points
 point_y	string	No		Vertical coordinates. Requires at least 2 points
 scale_value	string	No		Values corresponding to the coordinates, with the length consistent with

Name	Type	Required	Default	Remarks
				point_x and point_y
└─ direction_line	object	No		Direction line
└─ point_x	string	No		x1,x2
└─ point_y	string	No		y1,y2
└─ indicator_light_color	string	No		Indicator color, unique feature of the indicator status recognition algorithm
└─ indicator_light_status	string	No		Indicator status, unique feature of the indicator status recognition algorithm
└─ lanes	object[]	No		Lane line configuration
└─ lane_id	integer	No		Lane ID
└─ lane_type	string	No		Lane type: BusLane common lane EmergencyLane Emergence lane
└─ lane_area	object	No		Lane area, 4-6 points
└─ point_x	string	No		Horizontal coordinates (‰)
└─ point_y	string	No		Vertical coordinates (‰)
└─ direction	object	No		Lane direction, 2 points

Name	Type	Required	Default	Remarks
└─ point_x	string	No		Horizontal coordinates (%)
└─ point_y	string	No		Vertical coordinates (%)
└─ virtual_loop	object	No		Lane section (virtual coil), 4 points
└─ point_x	string	No		Horizontal coordinates (%)
└─ point_y	string	No		Vertical coordinates (%)
person_control_info	object	No		Person monitoring info
└─ alarm_threshold	integer	Yes		Alarm threshold (0-100)
└─ enable	integer	Yes		Enable or disable monitoring. 0: Disable 1: Enable
└─ control_type	integer	Yes		Monitoring type. 1: All 2: Match 3: Not match Currently, for non-face targets, the type must be All.
└─ person_lib_id	integer []	No		Person library ID
channel_id	integer	Yes		Channel ID
preset_id	integer	No		Preset ID (used when sending algorithms to PTZ camera presets)

Name	Type	Required	Default	Remarks
work_scene	integer	No		Working mode of traffic event tasks 0: Manual Mode 1: Configuration-Free Mode

Return Data

Name	Type	Required	Default	Remarks
code	integer	Yes		0: Success 400363: Failed to add intelligent analysis task 400382: Channel has enabled monitoring, unable to configure analysis task 400314: Channel does not exist 400323: Channel offline, unable to configure 400362: Parameter error when adding intelligent analysis task 400366: No drawn area 400377: Direction line error 400378: Time threshold cannot be negative 400379: Person limit cannot be negative 400380: Fluctuating number of people cannot be negative 400381: Background refresh interval cannot be negative 400376: the selected work clothes library has no successfully modeled work clothes 400369: Failed to modify intelligent analysis task 400370: Failed to call intelligent analysis task 400333: Failed to save arming time
msg	string	Yes		

Example

Request:

```
{  
  "smart_list": [  
    ...  
  ]  
}
```

```
{
  "algo_model": "WorkClothesAlarm",
  "workclothes_lib_list": [
    1
  ],
  "graphs": [
    {
      "graph_type": "polygon",
      "polygon": {
        "point_x": "50,9950,9950,50",
        "point_y": "50,50,9950,9950"
      }
    }
  ],
  "title": "No Work Clothes Detection",
  "id": 5,
  "paintType": "polygon"
}
],
"channel_id": 1
}
```

Response:

```
{
  "code": 0,
  "msg": "Succeed",
  "status_code": 0
}
```

6.2 Get Intelligent Analysis Task List

Basic Info

Path: /api/v2/smart/list

Method: POST

Interface description:

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks
channel_id	integer	Yes		Channel ID
preset_id	integer	No		Preset ID

Return Data

Name	Type	Required	Default	Remarks
code	number	Yes		0: Success 400314: Channel does not exist 400361: Failed to get intelligent analysis task list
data	object	No		
└ smart_list	object []	No		Intelligent analysis task list
└─ algo_model	string	Yes		Algorithm model. Max length: 64
└─ time_threshold	integer	No		Time threshold (second)
└─ person_num_limit	integer	No		Person number limit

Name	Type	Required	Default	Remarks
└─ back_time_threshold	integer	No		Background refresh interval (second)
└─ person_float_limit	integer	No		Fluctuating number of people
└─ line_dpc_enable	integer	No		Whether to enable tripwire people counting reset (VMS does not support this parameter) 1: enabled 0: disabled
└─ line_dpc_time	integer	No		Time when tripwire people counting results will be reset: Integer, in the range of 0-23 (VMS does not support this parameter).
└─ object_type	integer []	否		Type of intrusion detection target. 1: Pedestrian 2: Non-motor vehicle 3: Motor vehicle 5 Bear, 6 Wild Boar, 7 Tiger, 8 Horse, 9 Cattle, 10 Pig, 11 Sheep, 12 Cat, 13 Dog, 14 Leopard, 15 Wolf, 16 Deer, 17 Kangaroos, 99 Others
└─ max_person_object_height	integer	No		Max head & shoulder height for intrusion detection. Range: 0-1500 (VMS does

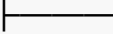
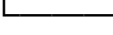
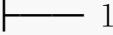
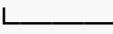
Name	Type	Required	Default	Remarks
				not support this parameter).
 max_person_object_width	integer	No		Max head & shoulder width for intrusion detection. Range: 0-1500 (VMS does not support this parameter).
 min_person_object_height	integer	No		Min head & shoulder height for intrusion detection. Range: 0-1500 (VMS does not support this parameter).
 min_person_object_width	integer	No		Min head & shoulder width for intrusion detection. Range: 0-1500 (VMS does not support this parameter).
 speed_info	object	No		Speeding parameter
 default_speed_limit	integer	No		Default speed limit (km/h)
 speed_limit_fluctuate	integer	No		Actual speed limit (percentage)
 speed_correction_infos	object []	No		Speed correction, three groups
 speed_correction	integer	No		Correction speed ratio
 speed_max	integer	No		Maximum correction speed
 speed_min	integer	No		Minimum correction speed

Name	Type	Required	Default	Remarks
└── congestion_info	Object	No		Congestion parameter
└── congestion_num	Integer	No		Congestion time
└── congestion_num	Integer	No		Number of congested vehicles
└ enable_work_clothes_alarm	integer	No		Illegal fueling Alarm mode 0: All people 1: Non-staff members
└── fuel_unloading_info	Object	No		Fuel unloading parameter
└── person_on_duty_overtime	Integer	No		Time when the person leaves the post.
└── stabilize_time	Integer	No		Fuel stabilization time
└── incident_model	string	No		Detection task
└ park_time	integer	No		Parking time
└ water_color	string []	No		Abnormal water color, multiple selections allowed White Black Red Yellow Blue Green;
└ high_water	integer	No		Highest water level
└ low_water	integer	No		Lowest water level
└ alarmed_coal_quantity	integer	No		Proportion of the belt width taken up by coal blocks

Name	Type	Required	Default	Remarks
				(in ten-thousandths percentage)
└─ coal_option	string []	No		Coal quantity alarm, multiple selections allowed Empty: No coal Few: Few coal Normal: Normal quantity Full: Overview
└─ flush_time	integer	No		Background refresh time
└─ dog_info	object []	No		Dot detection parameters
└─ dog_leash_length	integer	No		Leash length
└─ ship_option	string []	No		Ship type, multiple selections allowed All: All SandShip: Sandship PassengerShip: Passenger ship
└─ ship_type	string []	No		Ship type, multiple selections allowed. Supported by since version 2.3.6 99: Other ships 1: Sand barge detection 2: Cargo ship detection 3: Passenger ship detection
└─ rise_detect_mode	integer	No		Getting up detection mode

Name	Type	Required	Default	Remarks
				1: Leave 0: Enter
report_interval	integer	No		Traffic event reporting interval
feature_mode	integer	No		Whether uses the default base library images 1: Yes 0: No
fall_mode	integer	No		Fall detection scene type 0: General scene 1: Escalator scene
autotimer_info	object	No		Long queue detection parameter
auto_hour	integer	No		Hour
auto_minute	integer	No		Minute
auto_seconds	integer	No		Second
auto_direction	integer	No		Enable/disable automatic direction recognition
algo_model_mode	integer	No		Climbing detection (1: Head-Shoulder Mode, 2: Full-Body Mode) Fight detection (3: Indoor, 4: Outdoor)
identity_recognition_param	object []	No		Identity recognition parameters
condition_type	integer	Yes		Condition type: 1: Match All Conditions 2: Match any condition

Name	Type	Required	Default	Remarks
└── condition_details	object	Yes		Condition
└── regulation	integer	Yes		Rule: Rule 1: $\leq$ Rule 2: $\geq$ Rule 3: $\geq$ and $\leq$ Rule 4: $\geq$ or $\leq$
└── meter_reading_value1	integer	Yes		Value 1
└── meter_reading_value2	integer	Yes		Value 2
└── lib_id	integer	Yes		Work clothes library ID
└── workclothes_lib_list	integer []	No		Work clothes library ID
└ regulation	string	No		Rule 1: $\leq$ Rule 2: $\geq$ Rule 3: $\geq$ and $\leq$ Rule 4: $\geq$ or $\leq$
└ meter_reading_value1	number	No		Value 1
└ meter_reading_value2	number	No		Value 2
└ alarm_condition	integer	No		Switch position: 1: Open 2: Closed
└── graphs	object []	Yes		Area info (Note: the range of all the coordinates is 0~10000)
└── graph_type	string	Yes		Area type. Area: polygon Tripwire: line Mixed: blend
└── polygon	object	No		Area coordinates. When the area type is polygon or

Name	Type	Required	Default	Remarks
				blend, this field represents the coordinates of the drawn area.
 point_x	string	Yes		Horizontal coordinates. When the area type is polygon or blend, it is required to draw 3-6 points; format: x1, x2, x3, x4, x5, x6
 point_y	string	Yes		Vertical coordinates. When the area type is polygon or blend, it is required to draw 3-6 points; format: y1, y2, y3, y4, y5, y6
 line	object	No		Tripwire coordinates. When the area type is line or blend, this field represents the coordinates of the tripwire.
 point_x	string	Yes		Horizontal coordinates. When the area type is line or blend, it is required to draw 2 points; format: x1, x2
 point_y	string	Yes		Vertical coordinates. When the area type is line or blend, it

Name	Type	Required	Default	Remarks
				is required to draw 2 points; format: y1,y2
 scalevalue	object	No		
 point_x	string	No		Horizontal coordinates. Requires at least 2 points
 point_y	string	No		Vertical coordinates. Requires at least 2 points
 scale_value	string	No		Values corresponding to the coordinates, with the length consistent with point_x and point_y
 direction_line	object	No		Direction line
 fuel_task_type	String	No		Fuel unloading detection type
 graph_usage	string	No		ROI: Analyzing area NotROI: Shield area
 indicator_light_color	string	No		Indicator color, unique feature of the indicator status recognition algorithm
 indicator_light_status	string	No		Indicator status, unique feature of the indicator status recognition algorithm

Name	Type	Required	Default	Remarks
└─ lanes	object[]	No		Lane line configuration
└─ lane_id	integer	No		Lane ID
└─ lane_type	string	No		Lane type: BusLane: common lane EmergencyLane: emergency lane
└─ lane_area	object	No		Lane area, 4 to 6 points
└─ point_x	string	No		Horizontal coordinates (‰)
└─ point_y	string	No		Vertical coordinates (‰)
└─ direction	object	No		Lane direction, 2 points
└─ point_x	string	No		Horizontal coordinates (‰)
└─ point_y	string	No		Vertical coordinates (‰)
└─ virtual_loop	object	No		Lane section (virtual coil), 4 points
└─ point_x	string	No		Horizontal coordinates (‰)
└─ point_y	string	No		Vertical coordinates (‰)
└─ person_control_info	object	No		Person monitoring info
└─ alarm_threshold	string	Yes		Alarm threshold (0-100)
└─ enable	integer	Yes		Enable or disable monitoring.

Name	Type	Required	Default	Remarks
				0: Disable 1: Enable
└ control_type	integer	Yes		Monitoring type. 1: All 2: Match 3: Not match Currently, for non-face targets, the type must be All.
└ person_lib_id	integer []	No		Person library ID
└ work_scene	integer	No		Working mode of traffic event tasks 0: Manual Mode 1: Configuration-Free Mode
msg	string	Yes		

Example

Request:

```
{  
  "channel_id": 1  
}
```

Response:

```
{  
  "code": 0,  
  "msg": "Succeed",  
  "status_code": 0,  
  "data": {  
    "smart_list": [  
      {  
        "algo_model": "FallOverAlarm",  

```

```
"time_threshold": 5,
"graphs": [
  {
    "polygon": {
      "point_x": "50,9950,50",
      "point_y": "50,50,9950,9950"
    },
    "graph_type": "polygon"
  }
]
}
```

6.3 Get Algorithm Model List

Basic Info

Path: /api/v2/algo/list

Method: POST

Interface description:

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks	Other Info
------	------	----------	---------	---------	------------

Return Data

Name	Type	Required	Default	Remarks
code	number	No		0: Success 400365: Failed to get operator type list
data	object	No		
algo_model	string []	Yes		Algorithm model
msg	string	No		

Example

Request:

Response:

```
{
  "code": 0,
  "msg": "Succeed",
  "status_code": 0,
  "data": {
    "algo_model": [
      "ObjectIsRecognized",
      "FieldDetectorObjectsInside",
      "LineRuleData",
      "AreaRuleData",
      "AccessElevatorAlarm",
      "ObjectRemoved",
      "SafetyHelmetAlarm",
      "WorkClothesAlarm",
      "TelephoningAlarm",
      "NoMaskAlarm",
      "FallOverAlarm",
      "OffDutyDetectionAlarm",
      "SleepingDetectionAlarm"
    ]
  }
}
```

```
"ReflectiveClothesDetectionAlarm",  
"SmokingAlarm",  
"PlayMobilePhoneDetection",  
"FireDetection",  
"FumesAlarmBegin",  
"CrowdDensityCriticalAlarm",  
"AbnormalParkingDetection",  
"ChannelBlockageDetection"  
]  
}  
}
```

6.4 Get Extension Configuration

Basic Info

Path: /api/v2/extend/get

Method: POST

Interface description:

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks
channel_id	integer	No		Channel ID. All extension configuration info will be returned if this field is empty.

Return Data

Name	Type	Required	Default	Remarks
code	integer	Yes		
data	object []	Yes		
└ channel_id	integer	Yes		Channel ID
└ status	integer	Yes		Enabled/disabled status. 1: Enabled 2: Disabled. When disabled, the channel uses the default sensitivity and alarm interval.
└ smart_list	object []	Yes		
└ algo_model	string	Yes		Algorithm model. Max length: 64
└ sensitive	integer	No		Sensitivity (1-100)
└ report_rate	integer	No		Alarm interval (unit: second; when set to 0, each alarm will be reported once only)
msg	string	Yes		

Example

Request:

```
{  
  "channel_id": 1  
}
```

Response:

```
{  
  "code": 0,  
  "msg": "Succeed",  
}
```

```
"status_code": 0,
"data": [
  {
    "status": 0,
    "channel_id": 1,
    "smart_list": [
      {
        "algo_model": "FieldDetectorObjectsInside",
        "sensitive": 80
      },
      {
        "algo_model": "LineDetectorCrossed",
        "sensitive": 80
      },
      {
        "algo_model": "EnterArea",
        "sensitive": 80
      },
      {
        "algo_model": "LeaveArea",
        "sensitive": 80
      },
      {
        "algo_model": "ObjectRemoved",
        "sensitive": 80,
        "report_rate": 0
      },
      {
        "algo_model": "LineRuleData",
        "sensitive": 80
      },
      {
        "algo_model": "AreaRuleData",
        "sensitive": 80
      },
      {
        "algo_model": "AccessElevatorAlarm",
        "sensitive": 80
```

```
        "report_rate": 0
    },
    {
        "algo_model": "SmokingAlarm",
        "sensitive": 80,
        "report_rate": 0
    },
    {
        "algo_model": "SafetyHelmetAlarm",
        "sensitive": 80,
        "report_rate": 0
    },
    {
        "algo_model": "WorkClothesAlarm",
        "sensitive": 80,
        "report_rate": 0
    },
    {
        "algo_model": "TelephoningAlarm",
        "sensitive": 80,
        "report_rate": 0
    },
    {
        "algo_model": "NoMaskAlarm",
        "sensitive": 80,
        "report_rate": 0
    },
    {
        "algo_model": "FallOverAlarm",
        "sensitive": 80,
        "report_rate": 0
    },
    {
        "algo_model": "LongStayDetection",
        "report_rate": 0
    },
    {
        "algo_model": "FightDetectionAlarm",
```

```
        "sensitive": 80,  
        "report_rate": 0  
    },  
    {  
        "algo_model": "OffDutyDetectionAlarm",  
        "report_rate": 0  
    },  
    {  
        "algo_model": "SleepingDetectionAlarm",  
        "sensitive": 80,  
        "report_rate": 0  
    },  
    {  
        "algo_model": "ReflectiveClothesDetectionAla  
rm",  
        "sensitive": 80,  
        "report_rate": 0  
    },  
    {  
        "algo_model": "PlayMobilePhoneDetection",  
        "sensitive": 80,  
        "report_rate": 0  
    },  
    {  
        "algo_model": "FireDetection",  
        "sensitive": 80,  
        "report_rate": 0  
    },  
    {  
        "algo_model": "FumesAlarmBegin",  
        "sensitive": 80,  
        "report_rate": 0  
    },  
    {  
        "algo_model": "CrowdDensityCriticalAlarm",  
        "report_rate": 0  
    },  
    {
```

```
{
  "algo_model": "AbnormalParkingDetection",
  "report_rate": 0
},
{
  "algo_model": "ChannelBlockageDetection",
  "report_rate": 0
}
]
}
```

6.5 Modify Extension Configuration

Basic Info

Path: /api/v2/extend/update

Method: POST

Interface description:

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks
list	object []	Yes		
└ channel_id	integer	Yes		Channel ID
└ status	integer	Yes		Enabled/disabled status. 1: Enabled 2: Disabled.

Name	Type	Required	Default	Remarks
				When disabled, the channel uses the default sensitivity and alarm interval.
└ smart_list	object []	Yes		
└─ algo_model	string	Yes		Algorithm model. Max length: 64
└─ sensitive	integer	No		Sensitivity (1-100)
└─ report_rate	integer	No		Alarm interval (unit: second; when set to 0, each alarm will be reported once only)

Return Data

Name	Type	Required	Default	Remarks
code	integer	Yes		
msg	string	Yes		

Example

Request:

```
{
  "list": [
    {
      "channel_id": 1,
      "smart_list": [
        {
          "algo_model": "FieldDetectorObjectsInside",
          "sensitive": 80
        },
        {
          "algo_model": "ObjectRemoved",

```

```
        "sensitive": 80,  
        "report_rate": 0  
    },  
    {  
        "algo_model": "LineRuleData",  
        "sensitive": 80  
    },  
    {  
        "algo_model": "AreaRuleData",  
        "sensitive": 80  
    },  
    {  
        "algo_model": "AccessElevatorAlarm",  
        "sensitive": 80,  
        "report_rate": 0  
    },  
    {  
        "algo_model": "SmokingAlarm",  
        "sensitive": 80,  
        "report_rate": 0  
    },  
    {  
        "algo_model": "SafetyHelmetAlarm",  
        "sensitive": 80,  
        "report_rate": 0  
    },  
    {  
        "algo_model": "WorkClothesAlarm",  
        "sensitive": 80,  
        "report_rate": 0  
    },  
    {  
        "algo_model": "TelephoningAlarm",  
        "sensitive": 80,  
        "report_rate": 0  
    },  
    {  
        "algo_model": "NoMaskAlarm",
```

```
    "sensitive": 80,  
    "report_rate": 0  
  },  
  {  
    "algo_model": "FallOverAlarm",  
    "sensitive": 80,  
    "report_rate": 0  
  },  
  {  
    "algo_model": "OffDutyDetectionAlarm",  
    "report_rate": 0  
  },  
  {  
    "algo_model": "SleepingDetectionAlarm",  
    "sensitive": 80,  
    "report_rate": 0  
  },  
  {  
    "algo_model": "ReflectiveClothesDetectionAla  
rm",  
    "sensitive": 80,  
    "report_rate": 0  
  },  
  {  
    "algo_model": "PlayMobilePhoneDetection",  
    "sensitive": 80,  
    "report_rate": 0  
  },  
  {  
    "algo_model": "FireDetection",  
    "sensitive": 80,  
    "report_rate": 0  
  },  
  {  
    "algo_model": "FumesAlarmBegin",  
    "sensitive": 80,  
    "report_rate": 0  
  },  
}
```

```
{
  {
    "algo_model": "CrowdDensityCriticalAlarm",
    "report_rate": 0
  },
  {
    "algo_model": "AbnormalParkingDetection",
    "report_rate": 0
  },
  {
    "algo_model": "ChannelBlockageDetection",
    "report_rate": 0
  }
],
"status": 1
}
]
```

Response:

```
{
  "code": 0,
  "msg": "Succeed",
  "status_code": 0
}
```

6.6 Get Channel Parameter Configuration

Basic Info

Path: /api/v2/smart/params/get

Method: POST

Interface description:

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks
channel_id	integer	Yes		Channel ID

Return Data

Name	Type	Required	Default	Remarks
code	integer	Yes		
data	object	Yes		
└ alarm_video_param	object	Yes		Alarm video parameters
└─ status	integer	Yes		Whether alarm video is enabled (disabled by default) 1: enabled 0: disabled
└─ befor_alarmed_time	integer	Yes		Pre-alarm time (unit: second, range: 5-30)
└─ after_alarmed_time	integer	Yes		Post-alarm time (unit: second, range: 5-30)
└ status_recovery_param	object	Yes		Alarm reset parameter
└─ status	integer	Yes		Whether alarm reset is enabled (disabled by default) 1: enabled 0: disabled
└ face_param	object	Yes		Face parameters

Name	Type	Required	Default	Remarks
└─ min_width	string	Yes		Minimum face filter width. The default 10 is recommended.
└─ min_height	string	Yes		Minimum face filter height. The default 10 is recommended.
└─ prefer_last_time	integer	Yes		Speed priority parameter.
└─ interval_time	integer	Yes		Periodic selection parameter.
└─ filter_mode	integer	Yes		Face filter mode 0: Quality priority 1: Speed priority 2: Periodic selection 3: Quick report
msg	string	Yes		

Example

Request:

```
{  
  "channel_id": 1  
}
```

Response:

```
{  
  "code": 0,  
  "msg": "Succeed",  
  "status_code": 0,  
  "data": {  
    "alarm_video_param": {  
      "status": 0,  
      "befor_alarmed_time": 5,  
      "after_alarmed_time": 5  
    },  
  },  
}
```

```
"face_param": {
  "min_width": 30,
  "min_height": 30
},
"status_recovery_param": {
  "status": 0
}
}
```

6.7 Modify Channel Parameter Configuration

Basic Info

Path: /api/v2/smart/params/update

Method: POST

Interface description:

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks
channel_id	integer	Yes		Channel ID
alarm_video_param	object	No		Alarm video parameters
└ status	integer	Yes		Whether alarm video is enabled (disabled by default) 1: enabled 0: disabled

Name	Type	Required	Default	Remarks
└ befor_alarmed_time	integer	Yes		Pre-alarm time (unit: second, range: 5-30)
└ after_alarmed_time	integer	Yes		Post-alarm time (unit: second, range: 5-30)
status_recovery_param	object	No		Alarm reset parameter
└ status	integer	Yes		Whether alarm reset is enabled (disabled by default) 1: enabled 0: disabled
face_param	object	No		Face parameters
└ min_width	integer	Yes		Minimum face filter width. The default 10 is recommended.
└ min_height	integer	Yes		Minimum face filter height. The default 10 is recommended.
└ prefer_last_time	integer	Yes		Speed priority parameter.
└ interval_time	integer	Yes		Periodic selection parameter.
└ filter_mode	integer	Yes		Face filter mode 0: Quality priority 1: Speed priority 2: Periodic selection 3: Quick report

Return Data

Name	Type	Required	Default	Remarks
code	integer	Yes		
msg	string	Yes		

Example

Request:

```
{
  "channel_id": 1,
  "alarm_video_param": {
    "status": 0,
    "befor_alarmed_time": 5,
    "after_alarmed_time": 5
  },
  "status_recovery_param": {
    "status": 0
  },
  "face_param": {
    "min_width": 30,
    "min_height": 30
  }
}
```

Response:

```
{
  "code": 0,
  "msg": "Succeed",
  "status_code": 0
}
```

6.8 Task Activation/Deactivation Interface

Basic Info

Path: /api/v2/smart/task

Method: POST

Interface description:

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks
channel_id	integer	Yes		Channel ID
status	integer	Yes		1: Enabled 0: Disabled

Return Data

Name	Type	Required	Default	Remarks
code	integer	Yes		
msg	string	Yes		

Example

Request:

```
{
  "channel_id": 1,
  "status": 1
}
```

Response:

```
{
  "code": 0,
  "msg": "Succeed",
  "status_code": 0
}
```

6.9 Real-time Group Sequence Settings

Basic Info

Path: /api/v2/smart/cycle

Method: POST

Interface description: **Applicable to EIA only.**

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks
channel_id	integer	Yes		Channel ID
status	integer	Yes		1: Real-time 2: Group sequence

Return Data

Name	Type	Required	Default	Remarks
code	integer	Yes		
msg	string	Yes		

Example

Request:

```
{  
  "channel_id": 1,  
}
```

```
"status": 1  
}
```

Response:

```
{  
  "code": 0,  
  "msg": "Succeed",  
  "status_code": 0  
}
```

7 Work Clothes Library Management

7.1 Add Work Clothes Library

Basic Info

Path: /api/v2/workclotheslib/add

Method: POST

Interface description:

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks	Other Info
lib_name	string	Yes		Work clothes library name	Max length: 64 Min length: 1

Return Data

Name	Type	Required	Default	Remarks
code	integer	Yes		
data	object	Yes		
└ lib_id	integer	Yes		Work clothes library ID

Name	Type	Required	Default	Remarks
msg	string	Yes		

Example

Request:

```
{  
  "lib_name": "lib1"  
}
```

Response:

```
{  
  "code": 0,  
  "msg": "Succeed",  
  "status_code": 0,  
  "data": {  
    "lib_id": 2  
  }  
}
```

7.2 Modify Work Clothes Library

Basic Info

Path: /api/v2/workclotheslib/update

Method: POST

Interface description:

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks	Other Info
lib_id	integer	Yes		Work clothes library ID	
lib_name	string	Yes		Work clothes library name	Max length: 64 Min length: 1

Return Data

Name	Type	Required	Default	Remarks
code	integer	Yes		0: Success
msg	string	Yes		

Example

Request:

```
{  
  "lib_name": "lib2",  
  "lib_id": 2  
}
```

Response:

```
{  
  "code": 0,  
  "msg": "Succeed",  
  "status_code": 0  
}
```

7.3 Delete Work Clothes Library

Basic Info

Path: /api/v2/workclotheslib/delete

Method: POST

Interface description:

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks
lib_id	integer	Yes		Work clothes library ID

Return Data

Name	Type	Required	Default	Remarks
code	integer	Yes		
msg	string	Yes		

Example

Request:

```
{  
  "lib_id": 2  
}
```

Response:

```
{
  "code": 0,
  "msg": "Succeed",
  "status_code": 0
}
```

7.4 Get Work Clothes Library Info

Basic Info

Path: /api/v2/workclotheslib/list

Method: POST

Interface description:

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks
lib_name	string	No		Work clothes library name (supports fuzzy match)
page	integer	Yes		Current page
pagesize	integer	Yes		Number of rows per page

Return Data

Name	Type	Required	Default	Remarks
code	integer	Yes		
data	object	No		
└ workclothes_lib_list	object []	Yes		
└ workcloth_lib_list	object []	Yes		
└─ lib_id	integer	Yes		Work clothes library ID
└─ lib_name	string	Yes		Work clothes library name. Max length: 64
└─ create_time	integer	Yes		Creation time, 10-digit timestamp (unit: second)
└ total	integer	Yes		Total count
msg	string	Yes		

Example

Request:

```
{  
  "lib_name": "",  
  "page": 1,  
  "pagesize": 12  
}
```

Response:

```
{  
  "code": 0,  
  "msg": "Succeed",  
  "status_code": 0,  
  "data": {  
    "total": 1,  
  }  
}
```

```
"workcloth_lib_list": [  
  {  
    "lib_id": 1,  
    "lib_name": "default work clothes library",  
    "create_time": 1550108725  
  }  
]  
}  
}
```

8 Work Clothes Management

8.1 Add Work Clothes - Single

Basic Info

Path: /api/v2/workclothes/add

Method: POST

Interface description:

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks
lib_id	number	Yes		Work clothes library ID
image_base64	string	Yes		Work clothes image, Base64-encoded

Return Data

Name	Type	Required	Default	Remarks
code	integer	Yes		
data	object	No		
└ workclothes_id	integer	No		Work clothes ID
msg	string	Yes		

Example**Request:**

```
{  
  "lib_id": 1,  
  "image_base64": "/9j/4AAQSkZJRgABAQEAYABg"  
}
```

Response:

```
{  
  "code": 0,  
  "msg": "Succeed",  
  "status_code": 0,  
  "data": {  
    "workclothes_id": 2001  
  }  
}
```

8.2 Add Work Clothes - Batch

Basic Info

Path: /api/v2/workclothes/batchadd

Method: POST

Interface description:

Up to 50 allowed at a time.

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks
lib_id	integer	Yes		Work clothes library ID
image_base64	string []	Yes		Work clothes image, Base64-encoded

Return Data

Name	Type	Required	Default	Remarks
code	integer	Yes		
data	object	No		
└ workclothes_id	integer []	No		ID of successful work clothes
└ fail_list	object []	No		Failed image list
└─ index	integer	Yes		ID of failed image, starts from 0.
└─ reason	string	Yes		Failure cause. Max length: 64
msg	string	Yes		

Example

Request:

```
{
  "lib_id": 1,
  "image_base64": [
    "/9j/4AAQSkZJRgABAQEAYABg"
  ]
}
```

```
}
```

Response:

```
{
  "code": 0,
  "msg": "Succeed",
  "status_code": 0,
  "data": {
    "workclothes_id": [
      2001
    ],
    "fail_list": []
  }
}
```

8.3 Delete Work Clothes

Basic Info

Path: /api/v2/workclothes/delete**Method:** POST**Interface description:**

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks
lib_id	integer	Yes		Work clothes library ID

Name	Type	Required	Default	Remarks
workclothes_id_list	integer []	Yes		Work clothes ID

Return Data

Name	Type	Required	Default	Remarks
code	integer	Yes		
msg	string	Yes		

Example

Request:

```
{
  "lib_id": 1,
  "workclothes_id_list": [
    2001
  ]
}
```

Response:

```
{
  "code": 0,
  "msg": "Succeed",
  "status_code": 0
}
```

8.4 Get Work Clothes List

Basic Info

Path: /api/v2/workclothes/list

Method: POST

Interface description:

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks
lib_id	integer	Yes		Work clothes library ID
page	integer	Yes		Current page
pagesize	integer	Yes		Number of rows per page
model_status	integer	No		Modeling status. 999: All, 1: Not started, 2: In progress, 3: Succeeded 4: Failed

Return Data

Name	Type	Required	Default	Remarks
code	integer	Yes		
data	object	Yes		
└ total	integer	Yes		Total count
└ workclothes_list	object	Yes		
└ └ lib_id	number	Yes		Work clothes library ID
└ └ workclothes_id	number	Yes		Work clothes ID

Name	Type	Required	Default	Remarks
image_path	string	Yes		Relative path of image. Max length: 256
create_time	integer	Yes		Creation time, 10-digit timestamp, (unit: second)
model_status	number	No		Modeling status. 0: Not started 2: In progress, 3: Succeeded, 4: Failed
msg	string	Yes		

Example

Request:

```
{
  "lib_id": 1,
  "model_status": 999,
  "page": 1,
  "pagesize": 12
}
```

Response:

```
{
  "code": 0,
  "msg": "Succeed",
  "status_code": 0,
  "data": {
    "workclothes_list": [],
    "total": 0
  }
}
```

9 Device Info

9.1 Modify Device Info

Basic Info

Path: /api/v2/device/update

Method: POST

Interface description:

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks	Other Info
device_name	string	Yes		Device name	Max length: 64
device_code	integer	Yes		Device ID: 1-255.	Maximum value: 255
custom_code	string	No		Custom code	Max length: 64

Return Data

Name	Type	Required	Default	Remarks
code	integer	Yes		0: Success

Name	Type	Required	Default	Remarks
msg	string	Yes		

Example

Request:

```
{
  "device_name": "ECS-B501@8HA-SF",
  "device_code": 1
}
```

Response:

```
{
  "code": 0,
  "msg": "Succeed",
  "status_code": 0
}
```

9.2 Get Device Info

Basic Info

Path: /api/v2/device/get

Method: POST

Interface description:

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks	Other Info
------	------	----------	---------	---------	------------

Return Data

Name	Type	Required	Default	Remarks	Other Info
code	integer	Yes		0: Success	
msg	string	Yes			
data	object	Yes			
└ device_name	string	Yes		Device name	Max length: 64
└ device_model	string	Yes		Device model	Max length: 32
└ serial_number	string	Yes		Device serial number	Max length: 64
└ firmware_version	string	Yes		Firmware version	Max length: 128
└ release_time	integer	Yes		Version release time, time timestamp	
└ device_code	integer	Yes		Device ID, range: 1-255, default: 1	
└ custom_code	string	No		Custom code	Max length: 64

Example

Request:

Response:

```
{
```

```
"code": 0,  
"msg": "Succeed",  
"status_code": 0,  
"data": {  
  "device_name": "ECS-B501@8HA-SF",  
  "device_model": "ECS-B501@8HA-SF",  
  "serial_number": "210235C8TS5804696123",  
  "firmware_version": "ECS_B501_SF-B1105.2.05.240516",  
  "release_time": 1715796720,  
  "device_code": 1  
}  
}
```

10 Upgrade

10.1 Upload Package

Basic Info

Path: /api/v2/upgrade

Method: POST

Interface description:

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	multipart/form-data	Yes		

Body

Parameter	Type	Required	Example	Remarks
file	file	Yes		Upgrade package

Return Data

Name	Type	Required	Default	Remarks
code	integer	Yes		
msg	string	Yes		

Example

Request:

Response:

```
{
  "code": 0,
  "msg": "Succeed",
  "status_code": 0
}
```

10.2 Get Upgrade Status

Basic Info

Path: /api/v2/upgrade/status

Method: POST

Interface description:

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/x-www-form-urlencoded	Yes		

Return Data

Name	Type	Required	Default	Remarks
code	integer	Yes		
data	object	Yes		
└ status	integer	Yes		1: Transferring 2: Upgrading 3: Upgrade succeeded 4: Upgrade failed Note: The device will restart after the upgrade completes, and no response will be returned. Please wait patiently till the

Name	Type	Required	Default	Remarks
				device indicates success or failure.
msg	string	Yes		

Example

Request:

Response:

```
{
  "code": 0,
  "msg": "Succeed",
  "data": {
    "status": 2
  },
  "status_code": 0
}
```

11 Data Search

11.1 Searching Behavior Analysis Data

Basic Info

Path: /api/v2/search/behavior

Method: POST

Interface description:

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks
page	integer	Yes		Number of pages
pagesize	integer	Yes		Amount of data per page
channel_id	integer []	Yes		Channel ID
preset_id	integer []	No		Preset ID, only supported when channel_id is a single PTZ camera, retrieves specific preset data
algo_model	string[]	No		Algorithm model. All models will be searched when this field is empty.
start_time	integer	Yes		Start time, 10-digit timestamp (unit: second)

Name	Type	Required	Default	Remarks
end_time	integer	Yes		End time, 10-digit timestamp (unit: second)
compare_flag	integer	No		Whether there's a comparison. 1: All, 2: Compare Search all if absent.
object_type	integer[]	No		Target type. Query criteria for intrusion detection alarms: 1: Pedestrian 2: Motor vehicle 3: Non-motor vehicle 5: Bear 6: Wild boar
plate_no	string	No		Plate number
vehicle_status	integer	No		Parking space entry/exit status 1: Enter 2: Exit 999: All
gauge_status	string[]	No		Water level status: Low: Too low; High: Too high
coal_quantity	string[]	No		Coal quantity status: Empty: No coal Few: Few coal Normal: Normal quantity Full: Overflow
ship_type	integer[]	No		Ship type 99: Other ships 1: Sand barge detection 2: Cargo ship detection 3: Passenger ship detection

Return Data

Name	Type	Required	Default	Remarks	Other Info
code	integer	Yes		0: Success	
msg	string	Yes			
data	object	Yes			
└ total	integer	Yes		Total count	
└ list	object []	No			
└─ alarm_id	integer	Yes		Alarm ID	
└─ big_picture_url	string	Yes		URL of large alarm image	Max length: 256
└─ algo_model	string	Yes		Algorithm model	Max length: 128
└─ capture_time	integer	Yes		Alarm timestamp (unit: second)	
└─ channel_id	integer	Yes		Alarm channel	
└─ channel_name	string	Yes		Channel name	Max length: 64
└─ preset_id	integer	No		Preset ID	
└─ preset_name	string	No		Preset name	
└─ gauge_status	string	No		Water level status: Low: Too low High: Too high	
└─ coal_quantity	string	No		Coal quantity status. Empty: No coal Few: Few coal	

Name	Type	Required	Default	Remarks	Other Info
				Normal: Normal quantity Full: Overflow	
└─ circuit_breaker_status	string	No		Switch position: 1: Open, 2: Closed	
└─ pressure_gauge_scale	number	No		Pressure gauge reading	
└─ ocr_result	number	No		LED display number reading	
└─ indicator_light_info	object	No			
└─ indicator_light_color	string	No		Indicator color: Red, Green, Blue, White, Yellow	
└─ indicator_light_status	string	No		Indicator status: On/Off	
└─ video_url	string	No		Alarm video address	Max length: 256
└─ object_list	object []	Yes			
└─ speed	integer	Yes		Speed	
vehicle_status	integer	No		Parking space entry/exit status 1: Enter 2: Exit	
└─ object_type	integer	No		Target type 1: Pedestrian 2: Motor vehicle	

Name	Type	Required	Default	Remarks	Other Info
				3: Non-motor vehicle 6: Wild boar	
└── ship_type	integer	No		Ship type 99: Other ships 1: Sand barge detection 2: Cargo ship detection 3: Passenger ship detection	
└── plate_no	string	Yes		License plate number (only for illegal parking)	Max length: 16
└── compare_list	object []	No		Face comparison result	
└── base_image_id	integer	No		Face library photo ID	
└── base_image_url	string	No		URL of face library image	Max length: 256
└── name	string	No		Name	Max length: 64
└── certificate_no	string	No		ID No.	Max length: 64
└── similarity	integer	No		Matching degree (similarity): 1-100	

Name	Type	Required	Default	Remarks	Other Info
match_type	number	Yes		1: Match 2: Not match	

Example

Request:

```
{
  "page": 1,
  "pagesize": 10,
  "channel_id": [
    1,
    2,
    3,
    4,
    5,
    6,
    7,
    8
  ],
  "algo_model": [
    "FieldDetectorObjectsInside",
    "SafetyHelmetAlarm",
    "WorkClothesAlarm",
    "TelephoningAlarm",
    "NoMaskAlarm",
    "FallOverAlarm",
    "OffDutyDetectionAlarm",
    "SmokingAlarm",
    "SleepingDetectionAlarm",
    "ReflectiveClothesDetectionAlarm",
    "PlayMobilePhoneDetection",
    "FireDetection",
    "FumesAlarmBegin",
    "CrowdDensityCriticalAlarm",
    "AbnormalParkingDetection",
    "ChannelBlockageDetection",
  ]
}
```

```
        "AccessElevatorAlarm",
        "ObjectRemoved"
    ],
    "start_time": 1716480000,
    "end_time": 1716566399,
    "compare_flag": 1
}
```

Response:

```
{
    "code": 0,
    "msg": "Succeed",
    "status_code": 0,
    "data": {
        "total": 1210,
        "list": [
            {
                "alarm_id": 6318,
                "capture_time": 1716521787,
                "channel_id": 1,
                "channel_name": "Camera1",
                "algo_model": "SafetyHelmetAlarm",
                "big_picture_url": "/api/v2/smart/picture?Type=1
&Index=ubs_5_0_0_0_0_0_752_55_64647_1716521787&Size=222974",
                "object_list": [
                    {
                        "plate_no": ""
                    }
                ],
                "compare_list": [
                    {
                        "match_type": 1,
                        "base_image_id": 2005,
                        "name": "5",
                        "similarity": 95,
                        "certificate_no": "",

```

```

    "base_image_url": "/api/v2/smart/picture
?Type=3&Index=ubs_0_0_0_4_2_1716429337/Pic_1716429337015.jpg&Size=9999"
  }
]
}
]
}
}
```

11.2 Searching Face Snapshots Data

Basic Info

Path: /api/v2/search/facecap

Method: POST

Interface description:

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks	Other Info
page	integer	Yes		Number of pages	
pagesize	integer	Yes		Amount of data per page	
channel_id	integer []	Yes		Channel ID	

Name	Type	Required	Default	Remarks	Other Info
preset_id	integer []	No		Preset ID	
start_time	integer	Yes		Start time, 10-digit timestamp (unit: second)	
end_time	integer	Yes		End time, 10-digit timestamp (unit: second)	
faceattr	object	No			
└ gender	integer	No		Gender. 999: All 1: Male 2: Female 99: Unknown	
└ age	integer	No		Age. 999: All 5: Child 20: Teenager 40: Youth 60: Middle age 99: Senior	
└ glass	integer	No		Glasses. 999: All 1: No glasses 2: With glasses	
└ mask	integer	No		Mask. 999: All 1: No mask 2: With mask	
compare_flag	integer	No		Whether there's a comparison. 1: All, 2: Compare Search all if absent.	
name	string	No		Person name, supports fuzzy match Max length: 50 Min length: 10	Max length: 64
certificate_no	string	No		ID No., supports fuzzy match	Max length: 64

Name	Type	Required	Default	Remarks	Other Info
similarity	object	No		Matching degree (similarity): 0-100	
└ min	integer	Yes		Minimum value	
└ max	integer	Yes		Maximum value	
match_type	integer	No		Match result: 999: All 1: Match 2: Not match	

Return Data

Name	Type	Required	Default	Remarks	Other Info
code	integer	Yes		0: Success	
msg	string	Yes			
data	object	No			
└ total	integer	Yes		Total count	
└ list	object []	Yes			

Name	Type	Required	Default	Remarks	Other Info
└─ alarm_id	integer	Yes		Alarm ID	
└─ big_picture_url	string	Yes		URL of large alarm image	Max length: 256
└─ small_picture_url	string	Yes		URL of small alarm image	Max length: 256
└─ capture_time	integer	Yes		Alarm timestamp (unit: second)	
└─ channel_id	integer	Yes		Alarm channel	
└─ channel_name	string	Yes		Channel name	Max length: 64
└─ gender	integer	Yes		Gender: 1: Male 2: Female 99: Unknown	
└─ glass	integer	Yes		Glasses: 1: No glasses 2: With glasses	
└─ age	integer	Yes		Age 5: Child 20: Teenager 40: Youth 60: Middle age 99: Senior	

Name	Type	Required	Default	Remarks	Other Info
mask	integer	Yes		Mask: 1: No mask 2: With mask	
compare_list	object []	No			
base_image_id	integer	No		Face library photo ID	
base_image_url	string	No		URL of face library image	Max length: 256
name	string	No		Name	Max length: 64
similarity	integer	No		Matching degree (similarity): 1-100	
certificate_no	string	No		ID No.	Max length: 64
match_type	number	Yes		1: Match 2: Not match	

Example

Request:

```
{  
  "page": 1,  
  "pagesize": 10,  
  "channel_id": [  
    1,  
    2,  
    3,  
  ]  
}
```

```
4,  
5,  
6,  
7,  
8  
],  
"start_time": 1715702400,  
"end_time": 1716566399,  
"faceattr": {  
    "gender": 999,  
    "age": 999,  
    "glass": 999,  
    "mask": 999  
},  
"compare_flag": 1,  
"name": "",  
"match_type": 999,  
"certificate_no": "",  
"similarity": {  
    "min": 1,  
    "max": 100  
}  
}
```

Response:

```
{  
    "code": 0,  
    "msg": "Succeed",  
    "status_code": 0,  
    "data": {  
        "total": 2028,  
        "list": [  
            {  
                "alarm_id": 5109,  
                "capture_time": 1716464847,  
                "channel_id": 1,  
                "channel_name": "Camera1",
```

```
"algo_model": "ObjectIsRecognized",
"big_picture_url": "/api/v2/smart/picture?Type=1
&Index=ubs_5_0_0_0_0_0_751_75_40042_1716464847&Size=304742",
"small_picture_url": "/api/v2/smart/picture?Type
=2&Index=ubs_5_0_0_0_0_0_751_1_40041_1716464847&Size=3841",
"gender": 2,
"age": 40,
"glass": 1,
"mask": 1,
"compare_list": [
{
"match_type": 1,
"base_image_id": 2005,
"name": "5",
"similarity": 95,
"certificate_no": "",
"base_image_url": "/api/v2/smart/picture
?Type=3&Index=ubs_0_0_0_4_2_1716429337/Pic_1716429337015.jpg&Siz
e=9999"
}
]
}
]
}
}
```

11.3 Searching Behavior Analysis & Face Comparison Data

Basic Info

Path: /api/v2/search/behaviormatch

Method: POST

Interface description:

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks	Other Info
page	integer	Yes		Number of pages	
pagesize	integer	Yes		Amount of data per page	
channel_id	integer []	Yes		Channel ID	
preset_id	integer []	No		Preset ID	
start_time	integer	Yes		Start time, 10-digit timestamp (unit: second)	
end_time	integer	Yes		End time, 10-digit timestamp (unit: second)	
name	string	No		Person name	Max length: 64
certificate_no	string	No		ID No., supports fuzzy match	Max length: 64
algo_model	string []	No		Algorithm model. All models will be searched when this field is empty.	
similarity	object	No		Matching degree (similarity): 0-100	
└ min	integer	Yes		Minimum value	

Name	Type	Required	Default	Remarks	Other Info
└ max	integer	Yes		Maximum value	

Return Data

Name	Type	Required	Default	Remarks	Other Info
code	integer	Yes		0: Success	
msg	string	Yes			
data	object	No			
└ total	integer	Yes		Total count	
└ list	object []	Yes			
└─ alarm_id	integer	Yes		Alarm ID	
└─ big_picture_url	string	Yes		URL of large alarm image	Max length: 256
└─ algo_model	string	Yes		Alarm algorithm model	Max length: 128
└─ capture_time	integer	Yes		Alarm timestamp (unit: second)	
└─ channel_id	integer	Yes		Alarm channel	
└─ channel_name	string	Yes		Channel name	Max length: 64
└─ preset_id	integer	No		Preset ID	
└─ preset_name	string	No		Preset name	
└─ object_list	object []	No			
└─ base_image_id	integer	No		Face library photo ID	

Name	Type	Required	Default	Remarks	Other Info
base_image_url	string	No		URL of face library image	Max length: 256
name	string	No		Name	Max length: 64
certificate_no	string	No		ID No.	Max length: 64
similarity	integer	No		Matching degree (similarity): 1-100	
match_type	number	Yes		1: Match 2: Not match	

Example

Request:

```
{
  "page": 1,
  "pagesize": 10,
  "channel_id": [
    1,
    2,
    3,
    4,
    5,
    6,
    7,
    8
  ],
  "start_time": 1714579199,
  "end_time": 1716480000,
  "name": "",
  "match_type": 999,
  "certificate_no": "",
  "algo_model": [
    "FieldDetectorObjectsInside",
    "SafetyHelmetAlarm",
  ]
}
```

```
"WorkClothesAlarm",
"TelephoningAlarm",
"NoMaskAlarm",
"FallOverAlarm",
"OffDutyDetectionAlarm",
"SmokingAlarm",
"SleepingDetectionAlarm",
"ReflectiveClothesDetectionAlarm",
"PlayMobilePhoneDetection",
"FireDetection",
"FumesAlarmBegin",
"CrowdDensityCriticalAlarm",
"AbnormalParkingDetection",
"ChannelBlockageDetection",
"AccessElevatorAlarm",
"ObjectRemoved"
],
"similarity": {
  "min": 1,
  "max": 100
}
}
```

Response:

```
{
  "code": 0,
  "msg": "Succeed",
  "status_code": 0,
  "data": {
    "total": 131,
    "list": [
      {
        "alarm_id": 3741,
        "capture_time": 1716429656,
        "channel_id": 1,
        "channel_name": "Camera1",
        "algo_model": "NoMaskAlarm",
```

```
"big_picture_url": "/api/v2/smart/picture?Type=1
&Index=ubs_5_0_0_0_0_0_751_48_40622_1716429656&Size=192509",
"gender": 2,
"age": 40,
"glass": 1,
"mask": 1,
"object_list": [
  {
    "match_type": 1,
    "base_image_id": 2005,
    "name": "5",
    "similarity": 95,
    "certificate_no": "",
    "base_image_url": "/api/v2/smart/picture
?Type=3&Index=ubs_0_0_0_4_2_1716429337/Pic_1716429337015.jpg&Siz
e=9999"
  }
]
```

11.4 Searching Face Comparison Data

Basic Info

Path: /api/v2/search/facematch

Method: POST

Interface description:

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks	Other Info
page	integer	Yes		Number of pages	
pagesize	integer	Yes		Amount of data per page	
channel_id	integer []	Yes		Channel ID	
preset_id	integer []	No		Preset ID	
match_type	integer	No		Comparison result. 1: Match 2: Not match 99: All	
start_time	integer	Yes		Start time, 10-digit timestamp (unit: second)	
end_time	integer	Yes		End time, 10-digit timestamp (unit: second)	
name	string	No		Person name, supports fuzzy match Max length: 50 Min length: 10	Max length: 64
certificate_no	string	No		ID No., supports fuzzy match	Max length: 64
similarity	object	No		Matching degree (similarity): 0-100	
└ min	integer	Yes		Minimum value	
└ max	integer	Yes		Maximum value	
faceattr	object	No			

Name	Type	Required	Default	Remarks	Other Info
└ gender	integer	No		Gender. 999: All 1: Male 2: Female 3: Unknown	
└ age	integer	No		Age. 999: All 5: Child 20: Teenager 40: Youth 60: Middle age 99: Senior	
└ glass	string	No		Glasses. 999: All 1: No glasses 2: With glasses	
└ mask	string	No		Mask. 999: All 1: No mask 2: With mask	

Return Data

Name	Type	Required	Default	Remarks	Other Info
code	integer	Yes		0: Success	
msg	string	Yes			
data	object	No			
└ total	integer	Yes		Total count	
└ list	object []	Yes			
└ alarm_id	integer	Yes		Alarm ID	
└ big_picture_url	string	Yes		URL of large alarm image	Max length: 256
└ algo_model	string	Yes		Alarm algorithm model	Max length: 128

Name	Type	Required	Default	Remarks	Other Info
└─ capture_time	integer	Yes		Alarm timestamp (unit: second)	
└─ channel_id	integer	Yes		Alarm channel	
└─ channel_name	string	Yes		Channel name	Max length: 64
└─ preset_id	integer	No		Preset ID	
└─ preset_name	string	No		Preset name	
└─ object_list	object []	No			
└─ base_image_id	integer	No		Face library photo ID	
└─ base_image_url	string	No		URL of face library image	Max length: 256
└─ name	string	No		Name	Max length: 64
└─ similarity	integer	No		Matching degree (similarity): 1-100	
└─ certificate_no	string	No		ID No.	Max length: 64
└─ small_picture_url	string	Yes		URL of small alarm image	Max length: 256
└─ match_type	number	Yes		1: Match 2: Not match	

Example**Request:**

```
{
  "page": 1,
  "pagesize": 10,
  "channel_id": [
    1,
    2,
    3,
    4,
    5,
    6,
    7,
    8
  ],
  "start_time": 1714579199,
  "end_time": 1716480000,
  "name": "",
  "match_type": 999,
  "certificate_no": "",
  "similarity": {
    "min": 1,
    "max": 100
  }
}
```

Response:

```
{
  "code": 0,
  "msg": "Succeed",
  "status_code": 0,
  "data": {
    "total": 131,
    "list": [
      {
        "alarm_id": 3741,
        "capture_time": 1716429656,
        "channel_id": 1,
        "channel_name": "Camera1",

```

```
"algo_model": "ObjectIsRecognized",
"big_picture_url": "/api/v2/smart/picture?Type=1
&Index=ubs_5_0_0_0_0_0_751_48_40622_1716429656&Size=192509",
"gender": 2,
"age": 40,
"glass": 1,
"mask": 1,
"object_list": [
  {
    "small_picture_url": "/api/v2/smart/pict
ure?Type=2&Index=ubs_5_0_0_0_0_0_751_2_40620_1716429656&Size=478
3",
    "match_type": 1,
    "base_image_id": 2005,
    "name": "5",
    "similarity": 95,
    "certificate_no": "",
    "base_image_url": "/api/v2/smart/picture
?Type=3&Index=ubs_0_0_0_4_2_1716429337/Pic_1716429337015.jpg&Siz
e=9999"
  }
]
}
}
}
```

11.5 Tripwire People Counting Start

Basic Info

Path: /api/v2/search/line/start

Method: POST

Interface description:

Process of calling tripwire people counting interfaces

1. Call the Tripwire People Counting Start interface to pass in search criteria. After receiving the search criteria, the device starts retrieving the corresponding statistics and returns the search ID: `search_id`.
The search time periods cannot exceed 60; otherwise, the search will fail.
2. Call the Tripwire People Counting Progress interface with search ID. The device returns the current search progress. When the search progress is not 100, it is considered that the search is not completed, and the caller need to call the interface repeatedly to get the latest progress.
3. When the progress is 100, call the Tripwire People Counting Result interface. The device returns the search result.
4. When the Tripwire People Counting Stop interface is called, the device deletes the current task, and it can no longer get search results.

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks
num	integer	Yes		Number of channels to search
channels	integer []	No		Channel ID list
statistics_type	integer	Yes		Statistical type. 1: People entered, 2: People exited, 3: All
statistics_unit	integer	Yes		Statistical unit. 1: Minute, 2: Hour, 3: Day, 4: Month
begin	integer	Yes		Start time

Name	Type	Required	Default	Remarks
end	integer	Yes		End time

Return Data

Name	Type	Required	Default	Remarks	Other Info
code	integer	Yes			
msg	string	Yes			
data	object	Yes			
└ search_id	integer	Yes		Search ID	

Example

Request:

```
{
  "num": 8,
  "channels": [
    1,
    2,
    3,
    4,
    5,
    6,
    7,
    8
  ],
  "statistics_unit": 2,
  "statistics_type": 3,
  "begin": 1722481200,
  "end": 1722567600
}
```

Response:

```
{
  "code": 0,
  "msg": "Succeed",
  "status_code": 0,
  "data": {
    "search_id": 5
  }
}
```

11.6 Tripwire People Counting Progress

Basic Info

Path: /api/v2/search/line/progress

Method: POST

Interface description:

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks
search_id	integer	Yes		Search ID

Return Data

Name	Type	Required	Default	Remarks	Other Info
code	integer	Yes			
msg	string	Yes			
data	object	Yes			
└ percent	integer	Yes		Search progress, range: 0-100	

Example

Request:

```
{  
  "search_id": 5  
}
```

Response:

```
{  
  "code": 0,  
  "msg": "Succeed",  
  "status_code": 0,  
  "data": {  
    "percent": 100  
  }  
}
```

11.7 Tripwire People Counting Result

Basic Info

Path: /api/v2/search/line/result

Method: POST

Interface description:

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks
search_id	integer	Yes		Search ID

Return Data

Name	Type	Required	Default	Remarks	Other Info
code	integer	Yes			
msg	string	Yes			
data	object	Yes			
└ num	integer	Yes		Number of channels.	
└ statistics_data	object []	Yes		Tripwire people counting data of the channels	
└ id	integer	Yes		Channel ID	
└ time_section	integer	Yes		Number of time periods	
└ enter_count_list	integer []	Yes		Array of people	

Name	Type	Required	Default	Remarks	Other Info
				counting data -- number of people entered	
└─ exit_count_list	integer []	Yes		Array of people counting data -- number of people exited	
└─ search_status	integer	Yes		Search status. 1: Success; 2: Failure	

Example

Request:

```

{
  "search_id": 5
}
```

Response:

```

{
  "code": 0,
  "msg": "Succeed",
  "status_code": 0,
  "data": {
    "num": 1,
    "statistics_data": [{
      "id": 1,
      "time_section": 18,
      "search_status": 1,
    }]
```

```
        "enter_count_list": [0, 2, 10, 17, 16, 9, 10, 17, 16, 9, 10, 17, 16, 7],
        "exit_count_list": [0, 9, 17, 17, 9, 9, 17, 17, 9, 9, 17, 17, 9, 7, 17, 19, 9, 0]
    }],
    "enter_count": [0, 2, 10, 17, 16, 9, 10, 17, 16, 9, 10, 17, 16, 9, 10, 17, 16, 9, 10, 17, 16, 7],
    "exit_count": [0, 9, 17, 17, 9, 9, 17, 17, 9, 9, 17, 17, 9, 9, 17, 17, 9, 7, 17, 19, 9, 0],
    "present_count": [0, -7, -7, 0, 7, 0, -7, 0, 7, 0, -7, 0, 7, 0, -7, 0, 7, 2, -7, -2, 7, 7]
}
}
```

11.8 Tripwire People Counting Stop

Basic Info

Path: /api/v2/search/line/stop

Method: POST

Interface description:

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks
search_id	integer	Yes		Search ID

Return Data

Name	Type	Required	Default	Remarks	Other Info
code	integer	Yes			
msg	string	Yes			
data	object	Yes			

Example

Request:

```
{  
  "search_id": 5  
}
```

Response:

```
{  
  "code": 0,  
  "msg": "Succeed",  
  "status_code": 0,  
  "data": {}  
}
```

12 Platform Integration

12.1 Add/Modify Platform Integration Info

Basic Info

Path: /api/v2/docking/config/update

Method: POST

Interface description:

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks	Other Info
platform	object []	Yes		Platform Configuration	
└ id	integer	Yes		Platform ID. Up to 2 platforms allowed, 0	
└ version_code	string	Yes		Interface version. "2.0"	Max length: 128
└ enabled	integer	Yes		Interface version. 1: Enabled 0: Disabled	
└ url	string	Yes		Alarm reporting address	Max length: 256

Name	Type	Required	Default	Remarks	Other Info
└ picture_enable	integer	Yes		Enable or disable image upload. 1: Enable 0: Disable	
└ video_enable	integer	Yes		Enable or disable video upload. 1: Enable 0: Disable	
└ fuel_report_mode	integer	No		Fuel unloading reporting mode: 1: Separate reporting 2: Merged reporting	
└ pdf_enable	integer	No		Fuel unloading pdf reporting 1: Enabled 0: Disabled	
└ http_alive	integer	Yes		Enable or disable http keep-alive. 1: enable 0: disable	
└ alive_interval	integer	Yes		Keep-alive time interval, unit: second, range: 30-600	
└ channel_enable	integer	Yes		Enable or disable camera status reporting. 1: Enable 0: Disable	
└ alive_url	string	Yes		Keep-alive URL	
time_conf	object []	Yes		Upload time interval	
└ channel_id	integer	Yes		Channel ID	
└ interval	integer	Yes		Time interval, unit: minute, range: 0-10080	

Name	Type	Required	Default	Remarks	Other Info
L enable	integer	Yes		Enable or disable interval. 1: Enable 2: Disable	

Return Data

Name	Type	Required	Default	Remarks
code	integer	Yes		Error code
msg	string	Yes		Error info

Example

Request:

```
{
  "platform": [
    {
      "id": 1,
      "url": "http://217.1.1.188:8000",
      "enabled": 1,
      "picture_enable": 0,
      "video_enable": 0,
      "http_alive": 0,
      "alive_interval": 30,
      "channel_enable": 0,
      "version_code": "V2.0"
    },
    {
      "id": 2,
      "version_code": "20240423AIBOX",
      "enabled": 1,
      "url": "http://217.1.1.188:8002",
      "picture_enable": 0,
      "video_enable": 0,
    }
  ]
}
```

```
"http_alive": 0,  
"alive_interval": 30,  
"channel_enable": 0  
}  
],  
"time_conf": [  
  {  
    "channel_id": 1,  
    "enable": 0,  
    "interval": 100  
  },  
  {  
    "channel_id": 2,  
    "enable": 0,  
    "interval": 200  
  },  
  {  
    "channel_id": 3,  
    "enable": 0,  
    "interval": 300  
  },  
  {  
    "channel_id": 4,  
    "enable": 0,  
    "interval": 400  
  },  
  {  
    "channel_id": 5,  
    "enable": 1,  
    "interval": 500  
  },  
  {  
    "channel_id": 6,  
    "enable": 0,  
    "interval": 600  
  },  
  {  
    "channel_id": 7,
```

```
"enable": 1,
"interval": 700
},
{
  "channel_id": 8,
  "enable": 1,
  "interval": 800
}
]
```

Response:

```
{
  "code": 0,
  "msg": "Succeed",
  "status_code": 0
}
```

12.2 Get Platform Integration Info

Basic Info

Path: /api/v2/docking/config/get

Method: POST

Interface description:

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks
------	------	----------	---------	---------

Return Data

Name	Type	Required	Default	Remarks	Other Info
code	integer	Yes			
data	object	Yes			
└ time_conf	object []	Yes		Upload time configuration	
└─ channel_id	integer	Yes		Channel ID	
└─ interval	integer	Yes		Time interval, unit: second	
└─ enable	integer	Yes		Enable or disable interval. 1: Enable 0: Disable	
└─ channel_name	string	Yes		Channel name	Max length: 64
└ platform	object []	Yes			
└─ id	integer	Yes		Platform ID	
└─ version_code	string	Yes		Version number. Example: V1.0: AapiV1.0 V2.0: V2.0	Max length: 128
└─ enabled	integer	Yes		Enable or disable. 1: Enable 0: Disable	
└─ url	string	No		Alarm reporting address	Max length: 256

Name	Type	Required	Default	Remarks	Other Info
 picture_enable	integer	Yes		Enable or disable image upload. 1: Enable 0: Disable	
 video_enable	integer	Yes		Enable or disable video upload. 1: Enable 0: Disable	
 fuel_report_mode	integer	No		Fuel unloading reporting mode 1: Separate reporting 2: Merged reporting	
 pdf_enable	integer	No		Fuel unloading pdf reporting 1: Enabled 0: Disabled	
 http_alive	integer	Yes		Enable or disable http keep-alive. 1: enable 0: disable	
 alive_interval	integer	No		Keep-alive time interval, unit: minute	
 channel_enable	integer	Yes		Enable or disable camera status reporting. 1: Enable 0: Disable	
 alive_url	string	Yes		Keep-alive URL	
Msg	string	Yes			

Example

Request:

Response:

```
{
  "code": 0,
  "msg": "Succeed",
  "status_code": 0,
  "data": {
    "time_conf": [
      {
        "channel_id": 1,
        "enable": 0,
        "channel_name": "",
        "interval": 100
      },
      {
        "channel_id": 2,
        "enable": 0,
        "channel_name": "",
        "interval": 200
      },
      {
        "channel_id": 3,
        "enable": 0,
        "channel_name": "",
        "interval": 300
      },
      {
        "channel_id": 4,
        "enable": 0,
        "channel_name": "",
        "interval": 400
      },
      {
        "channel_id": 5,
        "enable": 1,
        "channel_name": "",
        "interval": 500
      },
      {
        "channel_id": 6,
```

```
"enable": 0,
"channel_name": "",
"interval": 600
},
{
  "channel_id": 7,
  "enable": 1,
  "channel_name": "",
  "interval": 700
},
{
  "channel_id": 8,
  "enable": 1,
  "channel_name": "",
  "interval": 800
}
],
"platform": [
  {
    "id": 1,
    "url": "http://217.1.1.188:8000",
    "enabled": 1,
    "picture_enable": 0,
    "video_enable": 0,
    "http_alive": 0,
    "alive_interval": 30,
    "channel_enable": 0,
    "version_code": "v2.0"
  }
]
}
}
```

12.3 Get Platform Integration Capability

Basic Info

Path: /api/v2/docking/ability/get

Method: POST

Interface description: **Applicable to EIA only.**

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks
------	------	----------	---------	---------

Return Data

Name	Type	Required	Default	Remarks	Other Info
code	integer	Yes			
data	object	Yes			
├ list	object []	Yes			
├ version_code	string	Yes		Version code	
├ version_name	string	Yes		Version name	

Example

Request:**Response:**

```
{
  "code": 0,
  "msg": "Succeed",
  "status_code": 0,
  "data": {
    "list": [
      {
        "version_code": "V1.0",
        "version_name": "1.0 interface"
      },
      {
        "version_code": "V2.0",
        "version_name": "2.0 interface"
      }
    ]
  }
}
```

13 Data Reporting

13.1 Alarm Data Reporting

Basic Info

Path: <http://ip:port/url> <https://ip:port/url> ws://ip:port/url

Method: POST

Interface description:

Alarm reporting: Supports HTTP, HTTPS, and WebSocket protocols, with consistent JSON data content.

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks	Other Info
type	integer	Yes		Reported contents 1: Behavior analysis alarm data 2: Alarm reset data 3: Face snapshot data 4: Face comparison data 5: Behavior comparison data	

Name	Type	Required	Default	Remarks	Other Info
				6: Keep-alive data 7: Channel change data	
event_id	string	Yes		Alarm ID	
has_result	integer	Yes		Whether there's comparison result 1: yes 0: no	
device_info	object	Yes		Device info	
└ device_code	integer	Yes		Device ID (1-255)	
└ device_name	string	Yes		Device name	Max length: 64
└ device_sn	string	Yes		Device serial number	Max length: 64
└ custom_code	string	No		Custom code	Max length: 64
channel_info	object	No		Channel info	
└ mode	integer	Yes		Reporting mode: 0: Channel info 1: Add channel 2: Modify channel 3: Delete channel 4: Channel status change	
└ ipc_sn	string	No		Camera's serial number	Max length: 64

Name	Type	Required	Default	Remarks	Other Info
└ ipc_addr	string	No		Camera IP address	
└ channel_id	integer	Yes		Channel ID	
└ channel_name	string	Yes		Channel name	Max length: 64
└ status	integer	No		Channel status. 1: online 2: offline	
└ custom_code	string	No		Custom code	Max length: 64
behaviour	object	No		Behavior analysis (alarm reset) data	
└ algo_model	string	Yes		Algorithm model	Max length: 128
└ image_base64	string	No		Image, Base64-encoded	
└ image_path	string	No		Relative path of image	Max length: 256
└ capture_time	integer	Yes		Snapshot time	
└ in_num	integer	No		Number of people entering the tripwire (instantaneous)	
└ out_num	integer	No		Number of people leaving the tripwire (instantaneous)	

Name	Type	Required	Default	Remarks	Other Info
└ area_num	integer	No		Number of people in the area	
└ gauge_scale	integer	No		Water level	
└ gauge_status	string	No		Water level status Normal: Normal Low: Very low High: Very high	
└ coal_quantity	string	No		Coal quantity status. Empty: No coal Few: Few coal Normal: Normal quantity Full: Overflow	
└ circuit_breaker_status	string	No		Switch position: 1: Open, 2: Closed	
└ pressure_gauge_scale	number	No		Pressure gauge reading	
└ ocr_result	number	No		LED display number reading	
└ indicator_light_info	object	No			
└ indicator_light_color	string	No		Indicator color: Red, Green, Blue, White, Yellow	
└ indicator_light_status	string	No		Indicator status: On/Off	
└ video_uuid	string	No		Alarm video uuid	Max length: 256

Name	Type	Required	Default	Remarks	Other Info
└ video_url	string	No		Relative path of alarm video	Max length: 256
└ capture_info	object []	No		Object info	
└ target_id	integer	No		Target ID	
└ point_x	string	No		Horizontal coordinates of the target-- "x0, x1"	
└ point_y	string	No		Vertical coordinates of the target-- "y0, y1"	
└ object_type	integer	No		Target type 1: Pedestrian 2: Motor vehicle 3: Non-motor vehicle 5: Bear 6: Wild boar 7 Tiger 8 Horse 9 Cattle 10 Pig 11 Sheep 12 Cat 13 Dog 14 Leopard 15 Wolf 16 Deer 17 Kangaroo	
└ speed	integer	No		Speed	
└ vehicle_status	integer	No		Parking space entry/exit status 1: Enter 2: Exit	

Name	Type	Required	Default	Remarks	Other Info
└─ ship_info	object	No		Ship detection information	
└─ ship_type	integer	No		Ship type 99: Other ships 1: Sand barge detection 2: Cargo ship detection 3: Passenger ship detection	
└─ vehicle_washing_info	object	No		Vehicle washing status information	
└─ washing_status	integer	No		Washing status 1: Vehicle unwashed 2: Vehicle bypassing to avoid washing 3: Insufficient vehicle washing water volume	
└─ plate_no	string	No		Plate number	
face	object	No		Face snapshot data	
└─ capture_time	integer	Yes		Snapshot time	
└─ point_x	string	No		Horizontal coordinates of the target-- "x0, x1"	
└─ point_y	string	No		Vertical coordinates of the target-- "y0, y1"	

Name	Type	Required	Default	Remarks	Other Info
└ orig_image_path	string	Yes		Relative path to the original face snapshot	Max length: 256
└ crop_image_path	string	Yes		Relative path to the cropped face snapshot	Max length: 256
└ orig_image_base64	string	No		Original image, Base64-encoded	
└ crop_image_base64	string	No		Cropped image, Base64-encoded	
└ video_uuid	string	No		Alarm video UUID (used for both snapshot and comparison videos)	
└ video_url	string	No		Relative path of the alarm video (used for both snapshot and comparison videos)	
└ video_rtsp	string	No		RTSP address of the alarm video (used for both snapshot and comparison videos)	
└ gender	integer	No		Gender 0: Unknown 1: Male 2: Female; 99: Unspecified	
└ glass	integer	No		Whether wearing glasses 1: No 2: Yes	

Name	Type	Required	Default	Remarks	Other Info
└ mask	integer	No		Whether wearing mask 1: No 2: Yes	
└ age	integer	No		Age 5: Child 20: Teenage 40: Yong Adult 60: Middle Age 99: Senior 0: Unknown	
traffic_parameter	integer	No		Traffic parameter data	
└ vehicle_volume	number	No		Traffic flow (vehicles/min) Floating point	
└ average_speed	number	No		Average speed (km/h) Floating point	
└ time_occupancy	number	No		Lane time occupancy rate (%) Floating point	
└ space_occupancy	number	No		Lane space occupancy rate (%) Floating point	
└ time_headway	number	No		Time headway (ms) Floating point	
└ space_headway	number	No		Space headway (m) Floating point	
└ queue_length	number	No		Queue length (m)	

Name	Type	Required	Default	Remarks	Other Info
				Floating point	
└ traffic_status	integer	No		Traffic status 1: Smooth 2: Congested 3: Blocked	
└ lane_id	integer	No		Lane ID	
compare_results	object []	No		Face comparison result (empty means comparison failed)	
└ target_id	integer	No		Target ID	
└ lib_id	integer	Yes		Person library ID	
└ lib_name	string	Yes		Person library name	Max length: 64
└ person_id	integer	Yes		Person ID	
└ person_name	string	Yes		Person name	Max length: 64
└ sex	integer	No		Gender 1: Male 2: Female 99: Unspecified	
└ email	string	No		Email	Max length: 64
└ tel	string	No		Mobile phone number	Max length: 32
└ certificate_type	integer	No		ID type. 1: ID card 2: passport 3:	

Name	Type	Required	Default	Remarks	Other Info
				driver's license 99: other	
└ certificate_no	string	No		ID No.	Max length: 64
└ image_path	string	Yes		Relative path to the face image in the library	Max length: 256
└ image_base64	string	No		Face image in library, Base64-encoded	
└ similarity	number	Yes		Matching degree (similarity)	

Return Data

Name	Type	Required	Default	Remarks
code	integer	Yes		
msg	string	Yes		

Example

Request:

Behavior analysis:

```
{  
  "event_id": "5503",  
  "has_result": 0,  
  "device_info": {  
    "device_code": 1,  
    "device_name": "ECS-B501@8HA-SF-NB",  
  }  
}
```

```
    "device_sn": "210235C8U43241000009"
  },
  "channel_info": {
    "mode": 0,
    "ipc_sn": "6H0BBA5PAG6189F",
    "ipc_addr": "217.1.1.51",
    "channel_id": 1,
    "channel_name": "1",
    "custom_code": "",
    "pull_sn": ""
  },
  "behaviour": {
    "algo_model": "FieldDetectorObjectsInside",
    "image_base64": "/9j/4AB//9k=",
    "image_path": "/api/v2/smart/picture?Type=1&Index=ubs_5_0_0_0_0_703_36_47158_1726650561&Size=146906",
    "capture_time": 1726650561,
    "capture_info": [
      {
        "point_x": "4515,5463",
        "point_y": "4240,8111",
        "plate_no": "",
        "target_id": 1,
        "object_type": 1
      }
    ]
  },
  "type": 1
}
```

Alarm Recovery:

```
{
  "event_id": "2",
  "has_result": 0,
  "device_info": {
    "device_code": 1,
    "device_name": "ECS-B50108HA-SY",
    "device_sn": "210235C7W43241000001"
  }
}
```

```
},
"channel_info": {
  "mode": 0,
  "ipc_sn": "6H0BBA5PAG6189F",
  "ipc_addr": "217.1.1.51",
  "channel_id": 1,
  "channel_name": "1",
  "custom_code": "",
  "pull_sn": ""
},
"behaviour": {
  "algo_model": "NoMaskAlarm",
  "capture_time": 1727679989
},
"type": 2
}
```

Face capture:

```
{
  "event_id": "5531",
  "has_result": 0,
  "device_info": {
    "device_code": 1,
    "device_name": "ECS-B501@8HA-SF-NB",
    "device_sn": "210235C8U43241000009"
  },
  "channel_info": {
    "mode": 0,
    "ipc_sn": "6H0BBA5PAG6189F",
    "ipc_addr": "217.1.1.51",
    "channel_id": 1,
    "channel_name": "1",
    "custom_code": "",
    "pull_sn": ""
  },
  "face": {
    "capture_time": 1726651071,
    "point_x": "5390,5552",
```

```
"point_y": "4425,4712",
"orig_image_path": "/api/v2/smart/picture?Type=1&Index=u
bs_5_0_0_0_0_0_636_37_24871_1726651072&Size=148045",
"crop_image_path": "/api/v2/smart/picture?Type=2&Index=u
bs_5_0_0_0_0_0_636_1_24870_1726651072&Size=1045",
"orig_image_base64": "/9j/FA2Q==",
"crop_image_base64": "/9j/4AAQ2Q=="
},
"type": 3
}
```

Face comparison:

```
{
  "event_id": "5544",
  "has_result": 1,
  "device_info": {
    "device_code": 1,
    "device_name": "ECS-B501@8HA-SF-NB",
    "device_sn": "210235C8U43241000009"
  },
  "channel_info": {
    "mode": 0,
    "ipc_sn": "6H0BBA5PAG6189F",
    "ipc_addr": "217.1.1.51",
    "channel_id": 1,
    "channel_name": "1",
    "custom_code": "",
    "pull_sn": ""
  },
  "compare_results": [
    {
      "lib_id": 2,
      "lib_name": "1",
      "person_id": 2001,
      "person_name": "99",
      "sex": 1,
      "certificate_type": 1,
      "certificate_no": "",

```

```
        "image_path": "/api/v2/smart/picture?Type=3&Index=ub
s_0_0_0_1_11_1726651189/Pic_1726651189398.jpg",
        "similarity": 73,
        "match_status": 1,
        "image_base64": "/9j/4+/2Q=="
    }
],
"face": {
    "capture_time": 1726651287,
    "point_x": "4750,5093",
    "point_y": "5527,6148",
    "orig_image_path": "/api/v2/smart/picture?Type=1&Index=u
bs_5_0_0_0_0_0_636_37_24379_1726651287&Size=148473",
    "crop_image_path": "/api/v2/smart/picture?Type=2&Index=u
bs_5_0_0_0_0_0_636_1_24378_1726651287&Size=2350",
    "orig_image_base64": "/9j/4KAP/Z",
    "crop_image_base64": "/9j/2Q=="
},
"type": 4
}
```

Behavior comparison:

```
{
    "event_id": "5550",
    "has_result": 1,
    "device_info": {
        "device_code": 1,
        "device_name": "ECS-B501@8HA-SF-NB",
        "device_sn": "210235C8U43241000009"
    },
    "channel_info": {
        "mode": 0,
        "ipc_sn": "6H0BBA5PAG6189F",
        "ipc_addr": "217.1.1.51",
        "channel_id": 1,
        "channel_name": "1",
        "custom_code": "",
        "pull_sn": ""
    }
}
```

```
},
"behaviour": {
  "algo_model": "NoMaskAlarm",
  "image_base64": "/9jk=",
  "image_path": "/api/v2/smart/picture?Type=1&Index=ubs_5_0_0_0_0_703_37_46215_1726651652&Size=149873",
  "capture_time": 1726651652,
  "capture_info": [
    {
      "point_x": "4927,5843",
      "point_y": "5111,7111",
      "plate_no": "",
      "target_id": 1
    }
  ]
},
"compare_results": [
  {
    "lib_id": 2,
    "lib_name": "1",
    "person_id": 2001,
    "person_name": "99",
    "similarity": 80,
    "sex": 1,
    "target_id": 1,
    "certificate_type": 1,
    "certificate_no": "",
    "image_path": "/api/v2/smart/picture?Type=3&Index=ubs_0_0_0_1_11_1726651189/Pic_1726651189398.jpg",
    "image_base64": "/9j/4AAQ2xt=="
  }
],
"type": 5
}

Keep-alive data:
{
  "device_info": {
```

```
"device_code": 1,
"device_name": "ECS-B501@8HA-SF-NB",
"device_sn": "210235C8U43241000009"
},
"type": 6
}
```

Channel change:

```
{
  "device_info": {
    "device_code": 1,
    "device_name": "ECS-B501@8HA-SF-NB",
    "device_sn": "210235C8U43241000009"
  },
  "channel_info": {
    "mode": 4,
    "ipc_sn": "2102412508WLL7010139",
    "ipc_addr": "217.1.1.56",
    "channel_id": 2,
    "channel_name": "222",
    "custom_code": "",
    "pull_sn": "",
    "status": 1
  },
  "type": 7
}
```

Response:

```
{
  "code": 0,
  "msg": "Succeed",
  "status_code": 0
}
```

13.2 Alarm Video Upload

Basic Info

Path: <http://ip:port/url>

Method: POST

Interface description:

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	multipart/form-data	Yes		

Body

Parameter	Parameter Type	Required	Example	Remarks
video_uuid	text	Yes		Unique identifier used to associate with the uploaded alarm data.
file	file	Yes		Alarm video stream

Return Data

Name	Type	Required	Default	Remarks
code	integer	Yes		Error code
msg	string	Yes		Error info

Example

Request:

Response:

```
{
```

```
"code": 0,  
"msg": "Succeed",  
"status_code": 0  
}
```

13.3 Evidence Chain File Reporting for Fuel Unloading Alarms

Basic Info

Path: /custom/router3

Method: POST

Interface description: **Applicable to AIBox only**

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	multipart/form-data	Yes		

Body

Parameter	Parameter Type	Required	Example	Remarks
video_uuid	text	Yes		Format is "Device Serial Number_Evidence Chain ID_Evidence Chain Status.pdf"
name	file	Yes		file

Return Data

Name	Type	Required	Default	Remarks
code	integer	Yes		Error Code
msg	string	Yes		Error info

Example

Request:

Http report fuel unloading evidence chain
Content-Type: multipart/form-data
Content-Disposition: form-data; name="file"; filename="device serial number_evidence chain ID_evidence chain status.pdf"

Example: 369002B012H217D00043_1_20220316154657_qualified
Device serial number: 369002B012H217D00043
Evidence chain ID: 1_20220316154657
Evidence chain status: qualified unqualified

Response:

```
{
  "code": 0,
  "msg": "Succeed"
}
```

13.4 Real-time Reporting of Fuel Unloading

Basic Info

Path: http://ip:port/url

Method: POST

Interface description: **Applicable to AIBox only**

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks	Other Info
oil_results	object []	No			
└ device_name	string	No		Device name	
└ device_addr0	string	No		IP address of the device's network port 1	

Name	Type	Required	Default	Remarks	Other Info
└ device_addr1	string	No		IP address of the device' s network port 2	
└ serial_number	string	No		Device' s serial number.	
└ IPC_serial_num	string	No		Camera' s serial number.	
└ IPC_addr	string	No		Camera' s IP address	
└ channel_id	string	No		Channel ID, used to identify a unique task ID	
└ channel_name	string	No		Channel name	
└ step_type	integer	No		Step type 23: Fuel tanker parking 24: Wheel chock placement 25: Safety device placement 26: Safety operation 27: Fuel hose connection 29: Personnel on duty 30: Post-unloading confirmation	
└ step_status	integer	No		Step status 1: qualified 2: not qualified	
└ step_id	integer	No		Step ID	

Name	Type	Required	Default	Remarks	Other Info
└ timestamp	integer	No		Collection time (unit: second, UTC format)	
└ big_image_url	string	No		URL of large fuel unloading analysis image	
└ big_image	object	No		Large fuel unloading analysis image	
└ Image	string	No		Base64-encoded image	
└ ImageType	string	No		Image type	
└ process_id	string	No		Evidence chain ID	
└ monitor_id	string	No		Channel ID	
└ process_status	string	No		Overall process status: Save Delete	
└ channel_serial_num	integer	No		Channel serial number	
└ soft_version	string	No		Firmware version	

Return Data

Name	Type	Required	Default	Remarks
code	integer	Yes		
msg	string	Yes		

Example

Request:

```
{
  "oil_results": [
    {
      "device_name": "ECS-B501@16HD-SF-NB",
      "device_addr0": "217.1.1.47",
      "device_addr1": "192.168.2.30",
      "serial_number": "210235C7W13248000012",
      "IPC_serial_num": "210235C7W13248000012_1",
      "IPC_addr": "217.1.1.138",
      "channel_id": "1",
      "channel_name": "111",
      "step_type": 30,
      "step_status": 1,
      "step_id": 30,
      "timestamp": 1733121842,
      "big_image_url": "/api/v2/smart/picture?Type=3&Index
=/fs_disk/ps/20241202/1_20241202144313/D1_20241202144313_TubingD
isconnection.jpg",
      "process_id": "1_20241202144313",
      "process_status": "Delete",
      "monitor_id": "111",
      "soft_version": "ECS_B501_SF-B1103.3.15.241129",
      "channel_serial_num": "1"
    }
  ]
}
```

Response:

```
{  
  "code": 0,  
  "msg": "Succeed"  
}
```

13.5 Non-realtime Reporting of Fuel Unloading

Basic Info

Path: http://ip:port/url

Method: POST

Interface description: **Applicable to AIBox only**

Request Parameter

Headers

Parameter	Value	Required	Example	Remarks
Content-Type	application/json	Yes		

Body

Name	Type	Required	Default	Remarks	Other Info
device_name	string	No		Device name	
device_addr0	string	No		IP address of the device's network port 1	

Name	Type	Required	Default	Remarks	Other Info
device_addr1	string	No		IP address of the device' s network port 2	
serial_number	string	No		Device' s serial number.	
IPC_serial_num	string	No		Camera' s serial number.	
IPC_addr	string	No		Camera' s IP address	
channel_id	string	No		Channel ID, used to identify a unique task ID	
channel_name	string	No		Channel name	
step_type	integer	No		Step type 23: Fuel tanker parking 24: Wheel chock placement 25: Safety device placement 26: Safety operation 27: Fuel hose connection 29: Personnel on duty 30: Post-unloading confirmation	
step_status	integer	No		Step status 1: qualified 2: not qualified	
step_id	integer	No		Step ID	

Name	Type	Required	Default	Remarks	Other Info
timestamp	integer	No		Collection time (unit: second, UTC format)	
big_image_url	string	No		URL of large fuel unloading analysis image	
big_image	object	No		Large fuel unloading analysis image	
Image	string	No		Base64-encoded image	
ImageType	string	No		Image type	
process_id	string	No		Evidence chain ID	
monitor_id	string	No		Channel ID	
process_status	string	No		Overall process status: Save Delete	
channel_serial_num	integer	No		Channel ID	
soft_version	string	No		Firmware version	

Return Data

Name	Type	Required	Default	Remarks
code	integer	Yes		
msg	string	Yes		

Example

Request:

```

{
  "device_name": "ECS-B501@16HD-SF-NB",
  "device_addr0": "217.1.1.47",
  "device_addr1": "192.168.2.30",
  "serial_number": "210235C7W13248000012",
  "IPC_serial_num": "210235C7W13248000012_1",
  "IPC_addr": "217.1.1.138",
  "channel_id": "1",
  "channel_name": "111",
  "step_type": 30,
  "step_status": 1,
  "step_id": 30,
  "timestamp": 1733122724,
  "big_image_url": "/api/v2/smart/picture?Type=3&Index=/fs_dis
k/ps/20241202/1_20241202145526/D1_20241202145526_TubingDisconnec
tion.jpg",
  "process_id": "1_20241202145526",
  "process_status": "Save",
  "monitor_id": "111",
  "soft_version": "ECS_B501_SF-B1103.3.15.241129",
  "channel_serial_num": "1"
}

```

Response:

```

{
  "code": 0,
  "msg": "Succeed"
}

```

Fuel unloading detection type

Charging Gun Not in Place Detection
No Fire Extinguisher Detection
Dump Truck Without Tarp Detection
Oil Leak Detection
Gas Leak Detection
Water Leak Detection
Test Paper Color Change Detection
No Safety Goggles Detection
No Safety Gloves Detection
No Dust/Gas Mask Detection

Special parameters:

Campus vehicle congestion detection: Vehicle Threshold, Congestion Duration

Dog detection: Capture Images of Large Dogs Only, Maximum Distance for Unleashed Dogs, and Permitted Dog Walking Hours.

14 Appendix

14.1 Error Code

● IA57XX@XX

0: Succeed

1: Common Error

2: Invalid Arguments

3: Not Authorized

4: Not Supported

5: Abnormal User Status

6: System Busy

1000 Authentication module

1000: Incorrect username or password

1001: User does not have permission

1002: The maximum number of user logins has been reached

1003: The user login ip has reached the upper limit.

1004: User login ip locked.

1100 Face lib management

1100 Face lib num has reached the upper limit

1101 The face lib name already exists

1200 Face management

1200: The number of faces has reached the max limit

1201: Wrong ID number type

1202: Image format error

1203: Image size is too large

1204: Face library is in use on another client.

1205: Insufficient UBS storage space for face library image -- no disk or insufficient space

1206: Image compression failed.

1207: Invalid file.

1208: The export task already exists.

1209: Invalid name.

1210: Invalid gender.

1211: Invalid date of birth.

1212: Invalid ID type.

1213: Invalid ID number.

1214: Invalid image path.

1215: Failed to import image.

1216: Invalid email address.
1217: Invalid mobile phone number.
1218: The import task already exists.

1300 Channel management

1400 Channel monitoring time

1500 Intelligence configuration
1500: Algo type does not support simultaneous config

1600 Work clothes lib management
1600: The workclothes lib name already exists
1601: Workclothes lib num has reached the upper limit

1700 Work clothes management
1700: The number of workclothes has reached the max limit
1701: Image size is too large
1702: Importing picture failed

1800 Device management
1800: Error in obtaining device information
1801: Error setting device information

1900 Upgrade
1900: Upgrade failure
1901: File verification failure
1902: The version number does not match
1903: The device type does not match
1904: Failed to apply for memory or other reasons
1905: Digital signature verification fails
1906: Failed to obtain upgrade progress

2000 Data search

2100 Platform connection

2200 Data reporting

14.2 Applicable Products

- IA57XX@XX
- ECS-X5X1
- ECS-50XX

14.3 Alarm Type

Parameter Description

Time unit: s

Parameter	Explanation	Range
sensitive	Sensitivity	1-100
report_rate	Alarm interval	≥ 0
direction_line	Direction line	/
time_threshold	Time threshold	≥ 0
person_num_limit	Person number limit	≥ 0
person_float_limit	Fluctuating number of people	≥ 0
back_time_threshold	Background refresh interval	≥ 0

ECS-X5X1-SF, ECS-50XX-SF:

Algorithm Type	Algorithm Name	Area Type	Sensitivity	Alarm Interval	Direction Line	Trigger Time	Person Number Limit	Fluctuating Number of People
SafetyHelmetAlarm	No Helmet Detection	polygon	✓	✓				
WorkClothesAlarm	No Work Clothes Detection	polygon	✓	✓				
TelephoningAlarm	Calling Detection	polygon	✓	✓				
SmokingAlarm	Smoking Detection	polygon	✓	✓				
SleepingDetectionAlarm	Sleep on Duty Detection	polygon	✓	✓		✓		
OffDutyDetectionAlarm	Absence Detection	polygon		✓		✓	✓	

Algorithm Type	Algorithm Name	Area Type	Sensitivity	Alarm Interval	Direction Line	Trigger Time	Person Number Limit	Fluctuating Number of People
ChannelBlockageDetection	Evacuation Route Obstruction	polygon		✓		✓		
ObjectRemoved	Object Left Behind	polygon	✓	✓		✓		
FieldDetectorObjectsInside	Intrusion Detection	polygon	✓					
AccessElevatorAlarm	Electric Moped Detection	polygon	✓	✓				
NoMaskAlarm	No Mask Detection	polygon	✓	✓				
FallOverAlarm	Fall Detection	polygon	✓	✓		✓		
CrowdDensityCriticalAlarm	Overcrowding Detection	polygon		✓		✓	✓	
ReflectiveClothesDetectionAlarm	No Reflective Clothing Detection	polygon	✓	✓				
AbnormalParkingDetection	Fire Lane Obstruction (Illegal Parking)	polygon		✓		✓		
FumesAlarmBegin	Smoke Detection	polygon	✓	✓				
PlayMobilePhoneDetection	Using Mobile Phone Detection	polygon	✓	✓		✓		
FireDetection	Fire Detection	polygon	✓	✓				
LongStayDetection	Long Stay Detection	polygon		✓		✓		
FightDetectionAlarm	Fight Detection	polygon	✓	✓				
LineDetectorCrossed	Cross Line Detection	polygon	✓					
EnterArea	Enter Area	polygon	✓					
LeaveArea	Leave Area	polygon	✓					
AreaRuleData	Area People Counting	polygon	✓					
LineRuleData	Tripwire People Counting	line	✓		✓			
ObjectIsRecognized	Face Detection	polygon						
NonMotorAbnormalParkingDetection	Illegally Parked Non-Motor	polygon		✓		✓		

Algorithm Type	Algorithm Name	Area Type	Sensitivity	Alarm Interval	Direction Line	Trigger Time	Person Number Limit	Fluctuating Number of People
	Vehicle Detection							
UncoveredTrashCanDetection	Trash Bin Open Detection	polygon	✓	✓		✓		
MouseDetect	Rat Detection	polygon	✓	✓				
BareSoilCoverDetection	UncoveredSoil	polygon	✓	✓				
DisorderStackingDetection	Improper Stacking of Materials	polygon	✓	✓		✓		
TrashOverflowingDetection	Trash Bin Overflow Detection	polygon	✓	✓		✓		
ExposedGarbageDetection	Exposed Trash Detection	polygon	✓	✓		✓		
PackedGarbageDetection	Bagged Trash Detection	polygon	✓	✓		✓		
ShirtlessDetection	Shirtless Detection	polygon	✓	✓				
ChefHatAlarm	No Chef Hat Detection	polygon	✓	✓				
ChefClothesDetection	No Chef Uniform Detection	polygon	✓	✓				
SafetyHarnessDetection	No Safety Harness Detection	polygon	✓	✓				
ClimbingDetectionAlarm	Climbing Detection	blend						
PeopleGathering	People Gathering	polygon					✓	
FastMoving	Quick Moving	polygon	✓	✓				
StayAloneDetection	Inadequate Worker Detection	polygon		✓		✓	✓	
KnifeStickDetection	Knife/Stick in Hand Detection	polygon	✓	✓				
UnwashedVehicleDetection	Unwashed Vehicle Detection	polygon				✓		
VehicleOverspeedDetection	Campus Vehicle Overspeed Detection	blend						

Algorithm Type	Algorithm Name	Area Type	Sensitivity	Alarm Interval	Direction Line	Trigger Time	Person Number Limit	Fluctuating Number of People
ForkliftOverspeedDetection	Forklift Overspeed Detection	blend						
NoSafetyBeltDetection	No Safety Belt Detection	polygon	✓	✓				
PresetMarkerDetection	Preset Marker Detection	polygon	✓	✓		✓		
GasCylinderDetection	Gas Cylinder Detection	polygon	✓	✓				
ChargingGunNotinPlace	Charging Gun Not in Place Detection	polygon		✓				
NoFireExtinguisherDetection	No Fire Extinguisher Detection	polygon	✓	✓		✓		
DumpTruckWithoutTarp	Dump Truck Without Tarp Detection	polygon	✓	✓				
OilLeakDetection	Oil Leak Detection	polygon		✓				
GasLeakDetection	Gas Leak Detection	polygon	✓	✓				
LiquidLeakDetection	Water Leak Detection	polygon	✓	✓		✓		
TestPaperColorChangeDetection	Test Paper Color Change Detection	polygon		✓				
NoSafetyGogglesDetection	No Safety Goggles Detection	polygon	✓	✓		✓		
NoSafetyGlovesDetection	No Safety Gloves Detection	polygon	✓	✓				
NoDustGasMaskDetection	No Dust/Gas Mask Detection	polygon	✓	✓		✓		
ExposedLongHairDetection	Exposed Long Hair Detection	polygon	✓	✓				
CampusEntranceExitLPC	Campus Entrance/Exit LPC	polygon						

Algorithm Type	Algorithm Name	Area Type	Sensitivity	Alarm Interval	Direction Line	Trigger Time	Person Number Limit	Fluctuating Number of People
CampusVehicleCongestionDetection	Campus Vehicle Congestion Detection	polygon	✓					
DogDetection	Dog Detection	polygon	✓	✓		✓		
FuelUnloadDetect	Fuel Unloading Detection	polygon						
Construction	Road Construction Detection	polygon						
ThrowingEvent	Litter Detection	polygon						
TrafficAccident	Traffic Accident Detection	polygon						
DriveSlowly	Vehicle Underspeed Detection	polygon						
DriveAway	Vehicle Driving Away Detection	polygon						
Fogging	Fog Detection	polygon						
AbnormalParkingDetection_HighSpeedEvent	Abnormal parking detection (traffic event)	polygon				✓		
AbnormalParkingDetection	Abnormal Parking Detection	polygon						
NonMotorVehicleIntrusionDetection	Non-Motor Vehicle Intrusion Detection	polygon						
OccupancyEmergencyLane	Emergency Lane Occupation Detection	polygon						
Pedestrian	Pedestrian Intrusion Detection	polygon						
Retrograde	Vehicle Wrong-Way Driving Detection	polygon						

Algorithm Type	Algorithm Name	Area Type	Sensitivity	Alarm Interval	Direction Line	Trigger Time	Person Number Limit	Fluctuating Number of People
SnowCover	Road Snow Accumulation Detection	polygon						
Congestion	Vehicle Congestion Detection	polygon						
VehicleEnterExitServiceStation	Vehicle Entering/Exiting Service/Charging Station	polygon						
ForkliftDetection	Forklift Detection	polygon						
EngineeringVehicleDetection	Construction Vehicle Detection	polygon						
IllegalAdditionOfBulkGasoline	Illegal Fueling Detection	polygon						
WildlifeIntrusionDetection	Animal Intrusion Detection	polygon	✓	✓				
FireOperationUnattended	Unattended Fire Detection	polygon	✓	✓				
SmokeAndFireDetectionEvent	Smoke and Fire Detection/Event	polygon						
RestrictedAreaFishingDetection	Restricted Area Fishing Detection	polygon	✓	✓				
WaterOutletDischargeDetection	Water Outlet Discharge Detection	polygon	✓	✓				
HandDetection	Hand Detection	polygon	✓	✓				
ElectricBicycleIntrusionDetection	Freight in Passenger Elevator Detection	polygon	✓	✓				
LongQueueDetection	Long Queue Detection	line	✓	✓		✓		
LightsLeftOnDetection	Light-On Detection	polygon	✓	✓		✓		

Algorithm Type	Algorithm Name	Area Type	Sensitivity	Alarm Interval	Direction Line	Trigger Time	Person Number Limit	Fluctuating Number of People
PedestrianAntiDirectionDetection	Pedestrian Reverse Motion Detection	polygon						
ReverseMotionOnEscalator	Reverse Motion on Escalator Detection	polygon	✓					
GunmanDetection	Gun Detection	polygon	✓					

Special parameters:

Campus vehicle congestion detection: Vehicle Threshold, Congestion Duration

Dog detection: Capture Images of Large Dogs Only, Maximum Distance for Unleashed Dogs, and Permitted Dog Walking Hours. Note: When AIBox configures/updates smart configurations or reports an alarm, the alarm type passed is DogDetection (Dog Detection). However, when retrieving data, the alarm type passed is DogWithoutLeashDetection (Unleashed Dog Detection), and DogWithoutLeashTimeDetection (Walking Dog During Prohibited Hours).

Unwashed vehicle detection: Note: When AIBox configures/updates smart configurations or reports an alarm, the alarm type passed is UnwashedVehicleDetection (Unwashed Vehicle Detection). However, when retrieving data, the alarm type passed is VehicleBypassingToAvoidWashing (Vehicle Bypassing to Avoid Washing), and InsufficientVehicleWashingWaterVolume (Insufficient Vehicle Washing Water Volume).

ECS-X5X1-SY:

Algorithm Type	Algorithm Name	Area Type	Sensitivity	Alarm Interval	Direction Line	Trigger Time	Person Number Limit	Fluctuating Number of People	Background Refresh Interval	Alarm Reset
KnifeStickDetection	Knife/Stick in Hand Detection	polygon	✓	✓						
SmokingAlarm	Smoking Detection	polygon	✓	✓						✓
SleepingDetectionAlarm	Sleep on Duty Detection	polygon	✓	✓		✓				✓
OffDutyDetectionAlarm	Absence Detection	polygon		✓		✓	✓			✓

Algorithm Type	Algorithm Name	Area Type	Sensitivity	Alarm Interval	Direction Line	Trigger Time	Person Number Limit	Fluctuating Number of People	Background Refresh Interval	Alarm Reset	Alarm Type
ClimbingDetectionAlarm	Climbing Detection	blend								✓	
FieldDetectorObjectsInside	Intrusion Detection	polygon	✓								✓
PeopleGathering	People Gathering	polygon				✓	✓	✓		✓	
FastMoving	Quick Moving	polygon	✓	✓						✓	
FallOverAlarm	Fall Detection	polygon	✓	✓		✓				✓	
FumesAlarmBegin	Smoke Detection	polygon	✓	✓						✓	
FireDetection	Fire Detection	polygon	✓	✓						✓	
LongStayDetection	Long Stay Detection	polygon		✓		✓				✓	
FightDetectionAlarm	Fight Detection	polygon	✓	✓						✓	
LineRuleData	Tripwire People Counting	line	✓		✓						
ObjectIsRecognized	Face Detection	polygon									

ECS-X5X1-VK:

Algorithm Type	Algorithm Name	Area Type	Sensitivity	Alarm Interval	Direction Line	Trigger Time	Person Number Limit	Fluctuating Number of People	Background Refresh Interval	Alarm Reset	Alarm Type
ChefHatAlarm	Not Chef Hat Detection	polygon	✓	✓							
SmokingAlarm	Smoking Detection	polygon	✓	✓						✓	
TelephoningAlarm	Calling Detection	polygon	✓	✓						✓	
NoMaskAlarm	No Mask Detection	polygon	✓	✓						✓	

Algorithm Type	Algorithm Name	Area Type	Sensitivity	Alarm Interval	Direction Line	Trigger Time	Person Number Limit	Fluctuating Number of People	Background Refresh Interval	Alarm Reset	Alarm Target Type
ChefClothesDetection	No Chef Uniform Detection	polygon	✓	✓							
UncoveredTrashCanDetection	Trash Bin Open Detection	polygon	✓	✓		✓					
ShirtlessDetection	Shirtless Detection	polygon	✓	✓							
MouseDetect	Rat Detection	polygon	✓	✓							
AreaRuleData	Area People Counting	polygon	✓								
LineRuleData	Tripwire People Counting	line	✓		✓						
ObjectIsRecognized	Face Detection	polygon									

ECS-X5X1-CS:

Algorithm Type	Algorithm Name	Area Type	Sensitivity	Alarm Interval	Direction Line	Trigger Time	Person Number Limit	Fluctuating Number of People	Detour Area	Background Refresh Interval
SafetyHelmetAlarm	No Helmet Detection	polygon	✓	✓						
WorkClothesAlarm	No Work Clothes Detection	polygon	✓	✓						
TelephoningAlarm	Calling Detection	polygon	✓	✓						
SmokingAlarm	Smoking Detection	polygon	✓	✓						
OffDutyDetectionAlarm	Absence Detection	polygon		✓		✓	✓			
FieldDetectorObjectsInside	Intrusion Detection	polygon	✓							
FallOverAlarm	Fall Detection	polygon	✓	✓		✓				

Algorithm Type	Algorithm Name	Area Type	Sensitivity	Alarm Interval	Direction Line	Trigger Time	Person Number Limit	Fluctuating Number of People	Detour Area	Background Refr Inter
CrowdDensityCriticalAlarm	Overcrowding Detection	polygon		✓		✓	✓			
ReflectiveClothesDetectionAlarm	No Reflective Clothing Detection	polygon	✓	✓						
AbnormalParkingDetection	Fire Lane Obstruction (Illegal Parking)	polygon		✓		✓				
ObjectIsRecognized	Face Detection	polygon								
FumesAlarmBegin	Smoke Detection	polygon	✓	✓						
FireDetection	Fire Detection	polygon	✓	✓						
NoMaskAlarm	No Mask Detection	polygon	✓	✓						
DisorderStackingDetection	Improper Stacking of Materials	polygon	✓	✓		✓				
BareSoilCoverDetection	Uncovered Soil	polygon	✓	✓						
UnwashedVehicleDetection	Unwashed Vehicle Detection	polygon				✓				
SafetyHarnessDetection	No Safety Harness Detection	polygon	✓	✓						

ECS-X5X1-GS:

Algorithm Type	Algorithm Name	Area Type	Sensitivity	Alarm Interval	Direction Line	Trigger Time	Person Number Limit	Fluctuating Number of People	Detour Area	Background Refr Inter
SafetyHelmetAlarm	No Helmet Detection	polygon	✓	✓						

Algorithm Type	Algorithm Name	Area Type	Sensitivity	Alarm Interval	Direction Line	Trigger Time	Person Number Limit	Fluctuating Number of People	Detour Area	Background Refr Inter
WorkClothesAlarm	No Work Clothes Detection	polygon	✓	✓						
TelephoningAlarm	Calling Detection	polygon	✓	✓						
SmokingAlarm	Smoking Detection	polygon	✓	✓						
SleepingDetectionAlarm	Sleep on duty detection	polygon	✓	✓		✓				
OffDutyDetectionAlarm	Absence Detection	polygon		✓		✓	✓			
ChannelBlockageDetection	Evacuation Route Obstruction	polygon		✓		✓				
ObjectRemoved	Object Removed	polygon	✓	✓		✓			✓	
FieldDetectorObjectsInside	Intrusion Detection	polygon	✓							✓
AccessElevatorAlarm	Elevator Entrance Detection	polygon	✓	✓						
NoMaskAlarm	No Mask Detection	polygon	✓	✓						
FallOverAlarm	Fall Detection	polygon	✓	✓		✓				
CrowdDensityCriticalAlarm	Overcrowding Detection	polygon		✓		✓	✓			
ReflectiveClothesDetectionAlarm	No Reflective Clothing Detection	polygon	✓	✓						
AbnormalParkingDetection	Fire Lane Obstruction (Illegal Parking)	polygon		✓		✓				
FumesAlarmBegin	Smoke Detection	polygon	✓	✓						

Algorithm Type	Algorithm Name	Area Type	Sensitivity	Alarm Interval	Direction Line	Trigger Time	Person Number Limit	Fluctuating Number of People	Detour Area	Background Refresh Interval
PlayMobilePhoneDetection	Using MobilePhone Detection	polygon	✓	✓		✓				
FireDetection	Fire Detection	polygon	✓	✓						
SurfaceWaterDetection	Ground Water Accumulation Detection	polygon	✓	✓		✓				
MouseDetect	Rat detection	polygon	✓	✓						
LiquidLeakDetection	Water Leak Detection	polygon	✓	✓		✓				
AreaRuleData	Area People Counting	polygon	✓							
LineRuleData	Tripwire People Counting	line	✓		✓					
ObjectIsRecognized	Face Detection	polygon								

IA57XX@PI:

Algorithm Type	Algorithm Name	Area Type	Sensitivity	Alarm Interval	Direction Line	Trigger Time	Person Number Limit	Fluctuating Number of People	Background Refresh Interval	Alarm Reset
LineRuleData	Tripwire People Counting	line	✓		✓					
AbnormalParkingDetection	Fire Lane Obstruction (Illegal Parking)	polygon		✓		✓				✓
TelephoningAlarm	Calling Detection	polygon	✓	✓						✓

Algorithm Type	Algorithm Name	Area Type	Sensitivity	Alarm Interval	Direction Line	Trigger Time	Person Number Limit	Fluctuating Number of People	Background Refresh Interval	Alarm Reset
FightDetectionAlarm	Fight Detection	polygon	✓	✓						✓
AccessElevatorAlarm	Electric Moped Detection	polygon	✓	✓						✓
FallOverAlarm	Fall Detection	polygon	✓	✓		✓				✓
LongStayDetection	Long Stay Detection	polygon		✓		✓				✓
StayAloneDetection	Inadequate Worker Detection	polygon		✓		✓	✓			✓
FastMoving	Quick Moving	polygon	✓	✓						✓
OffDutyDetectionAlarm	Absence Detection	polygon		✓		✓	✓			✓
FireDetection	Fire Alarm	polygon	✓	✓						✓
ClimbingDetectionAlarm	Climbing Detection	blend								✓
AreaRuleData	Area People Counting	polygon	✓							
FieldDetectorObjectsInside	Intrusion Detection	polygon	✓							✓
PeopleGathering	People Gathering	polygon					✓			
CrowdDensityCriticalAlarm	Overcrowding Detection	polygon		✓		✓	✓			✓
SleepingDetectionAlarm	Sleep on Duty Detection	polygon	✓	✓		✓				✓
ChannelBlockageDetection	Evacuation Route Obstruction	polygon		✓		✓				✓

Algorithm Type	Algorithm Name	Area Type	Sensitivity	Alarm Interval	Direction Line	Trigger Time	Person Number Limit	Fluctuating Number of People	Background Refresh Interval	Alarm Reset
PlayMobilePhoneDetection	Using Mobile Phone Detection	polygon	✓	✓		✓				✓
WorkClothesAlarm	No Work Clothes Detection	polygon	✓	✓						✓
SafetyHelmetAlarm	No Helmet Detection	polygon	✓	✓						✓
NoMaskAlarm	No Mask Detection	polygon	✓	✓						✓
SmokingAlarm	Smoking Detection	polygon	✓	✓						✓
FumesAlarmBegin	Smoke Detection	polygon	✓	✓						✓
LineDetectorCrossed	Cross Line Detection	line	✓		✓					
PresetMarkerDetection	Preset Marker Detection	polygon	✓	✓		✓				
ExposedGarbageDetection	Exposed Garbage Detection	polygon	✓	✓		✓				
ChargingGunNotinPlace	Charging Gun Not in Place	polygon	✓	✓						
PackedGarbageDetection	Packed Garbage Detection	polygon	✓	✓		✓				
WildlifeIntrusionDetection	Animal Intrusion Detection	polygon	✓	✓						
TrashOverflowingDetection	Trash Bin Overflow	polygon	✓	✓		✓				
GasLeakDetection	Gas Leak Detection	polygon	✓	✓						
LiquidLeakDetection	Water Leak Detection	polygon	✓	✓		✓				

Algorithm Type	Algorithm Name	Area Type	Sensitivity	Alarm Interval	Direction Line	Trigger Time	Person Number Limit	Fluctuating Number of People	Background Refresh Interval	Alarm Reset
OilLeakDetection	Oil Leakage Detection	polygon		✓						
BareSoilCoverDetection	Uncovered Soil Detection	polygon	✓	✓						
GasCylinderDetection	Gas Cylinder Detection	polygon	✓	✓						
NoFireExtinguisherDetection	No Fire Extinguisher Detection	polygon	✓	✓		✓				
DogDetection	Dog Detection	polygon	✓	✓		✓				
TestPaperColorChangeDetection	Test Paper Color Change Detection	polygon		✓						
MouseDetect	Rat Detection	polygon	✓	✓						
UncoveredTrashCanDetection	Trash Bin Open Detection	polygon	✓	✓		✓				
DisorderStackingDetection	Improper Stacking of Materials	polygon	✓	✓		✓				
ObjectRemoved	Object Removed	polygon	✓	✓		✓			✓	✓
DumpTruckWithoutTarp	Dump Truck Without Tarp Detection	polygon	✓	✓						
ExposedLongHairDetection	Exposed Long Hair Detection	polygon	✓	✓						
CrowdDispersion	Crowd Dispersion	polygon				✓		✓		
ReflectiveClothesDetectionAlarm	No Reflective	polygon	✓	✓						

Algorithm Type	Algorithm Name	Area Type	Sensitivity	Alarm Interval	Direction Line	Trigger Time	Person Number Limit	Fluctuating Number of People	Background Refresh Interval	Alarm Reset
	Clothing Detection									
NoSafetyBeltDetection	No Safety Belt Detection	polygon	√	√						
PedestrianAntiDirectionDetection	Pedestrian Reverse Motion Detection	polygon								
ChefHatAlarm	No Chef Hat Detection	polygon	√	√						
KnifeStickDetection	Knife/Stick in Hand Detection	polygon	√	√						
ShirtlessDetection	Shirtless Detection	polygon	√	√						
IndividualAloneDetection	Individual Alone Detection	polygon	√	√						
FareEvasionDetection	Fare Evasion Detection	polygon	√							
NoDustGasMaskDetection	No Dust/Gas Mask Detection	polygon	√	√		√				
NoSafetyGlovesDetection	No Safety Gloves Detection	polygon	√	√						
SafetyHarnessDetection	No Safety Harness Detection	polygon	√	√						
PassingThingsThroughFenceDetection	Passing Things Through Fence Detection	polygon	√							

Algorithm Type	Algorithm Name	Area Type	Sensitivity	Alarm Interval	Direction Line	Trigger Time	Person Number Limit	Fluctuating Number of People	Background Refresh Interval	Alarm Reset
PersonGetup	Getting Up Detection	polygon		√						
ChefClothesDetection	No Chef Uniform Detection	polygon	√	√						
NoSafetyGogglesDetection	No Safety Goggles Detection	polygon	√	√		√				
ForkliftOverspeedDetection	Forklift Overspeed Detection	polygon								
EngineeringVehicleDetection	Construction Vehicle Detection	polygon	√	√						
ForkliftDetection	Forklift Detection	polygon	√			√				
IllegalAdditionOfBulkGasoline	Illegal Fueling Detection	polygon	√							
VehicleEnterExitServiceStation	Vehicle Entering/Exiting Service/Charging Station	polygon								
VehicleOverspeedDetection	Campus Vehicle Overspeed Detection	polygon								
UnwashedVehicleDetection	Unwashed Vehicle Detection	polygon				√				
CampusVehicleCongestionDetection	Campus Vehicle Congestion Detection	polygon	√							

Algorithm Type	Algorithm Name	Area Type	Sensitivity	Alarm Interval	Direction Line	Trigger Time	Person Number Limit	Fluctuating Number of People	Background Refresh Interval	Alarm Reset
CampusEntranceExitLPC	Campus Entrance/Exit LPC	polygon								
NonMotorAbnormalParkingDetection	Illegally Parked Non-Motor Vehicle	polygon	√	√						
DriveSlowly	Vehicle Underspeed Detection	polygon	√							
Retrograde	Vehicle Wrong-Way Driving	polygon								
DriveAway	Vehicle Driving Away Detection	polygon								
Congestion	Vehicle Congestion Detection	polygon	√							
Construction	Road Construction Detection	polygon								
ThrowingEvent	Litter Detection	polygon	√							
SnowCover	Road Snow Accumulation Detection	polygon								
TrafficAccident	Traffic Accident Detection	polygon								
NonMotorVehicleIntrusionDetection	Non-Motor Vehicle Intrusion Detection	polygon								
Fogging	Fog Detection	polygon	√							

Algorithm Type	Algorithm Name	Area Type	Sensitivity	Alarm Interval	Direction Line	Trigger Time	Person Number Limit	Fluctuating Number of People	Background Refresh Interval	Alarm Reset
Pedestrian	Pedestrian Intrusion Detection	polygon								
AbnormalParkingDetection_HighSpeedEvent	Abnormal Parking Detection(traffic event)	polygon								
OccupancyEmergencyLane	Emergency Lane Occupation Detection	polygon								
OutsideTheStoreBusinessDetection	Out-of-Store Object	polygon	√	√		√				
ShipDetection	Ship Detection	polygon	√			√				
SurfaceWaterDetection	Ground Water Accumulation Detection	polygon	√	√		√				
SharedBicyclesRandomParkingDetection	Illegally Parked Shared Bicycle	polygon	√	√		√				
FloatingObjOnRiverDetection	Floating Debris in River	polygon	√	√						
WaterOutletDischargeDetection	Water Outlet Discharge Detection	polygon	√	√						
DirtyRoadDetection	Unclean Road Surface	polygon	√	√		√				
SetFileOnTheRoadDetection	Burning at Intersection	polygon	√	√						
RestrictedAreaSwimmingDetection	Restricted Area Swimming Detection	polygon	√							

Algorithm Type	Algorithm Name	Area Type	Sensitivity	Alarm Interval	Direction Line	Trigger Time	Person Number Limit	Fluctuating Number of People	Background Refresh Interval	Alarm Reset
RestrictedAreaFishingDetection	Restricted Area Fishing Detection	polygon	√	√						
WaterGaugeDetection	Water Level Detection	polygon		√						
StreetVendorDetection	Peddler Detection	polygon	√	√		√				
WasteTruckDetection	Dump Truck Detection	polygon	√	√						
AbnormalWaterColorDetection	Abnormal Water Color Detection	polygon	√	√						
IllegalUseOfAnUmbrellaDetection	Illegal Umbrella Shelter	polygon	√	√		√				
RoadsideStallBusinessDetection	Roadside Booth	polygon	√	√		√				
IllegalBillboardsDetection	Illegal Billboard	polygon	√	√						
DryingOnTheStreetDetection	Hanging Clothes Along Street	polygon	√	√						
BeltIdlingDetection	Conveyor Belt Empty Load Detection	polygon				√				
OversizedCoalDetection	Oversized Coal Detection	polygon		√						
BeltDeviationDetection	Conveyor Belt Deviation Detection	polygon		√						
CoalQuantityDetection	Coal Quantity Detection	polygon				√				

Algorithm Type	Algorithm Name	Area Type	Sensitivity	Alarm Interval	Direction Line	Trigger Time	Person Number Limit	Fluctuating Number of People	Background Refresh Interval	Alarm Reset
NotWearRubberShoesDetection	No Rubber Shoes Detection	polygon		√		√				
BeltRunningStateDetection	Conveyor Belt Start/Stop Detection	polygon								
BeltForeignObjectsDetection	Foreign Object on Conveyor Belt Detection	polygon		√						
NotWearCapLampDetection	No Headlamp Detection	polygon		√		√				
BeltPassThroughDetection	Cross Over Conveyor Belt Detection	polygon	√							√
NotWearSelfRescueBagDetection	No Self-Rescuer Detection	polygon		√		√				
TrafficParameter	Traffic Parameters									
FireOperationUnattended	Unattended Fire Detection	Polygon	√	√						
FreightInPassengerElevator	Freight In Passenger Elevator Detection	Polygon	√	√						
OnDutyPersonnelRatio	On-duty Personnel Ratio Detection	polygon	√	√		√				

Special parameters:

Getting Up Detection: Getting Up Detection Mode

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Dog Detection: Leash Length

Forklift Overspeed: Default Speed Limit, Speed Limit Ratio of Speeding, Speed Correction

Forklift Detection: Max. Motor Vehicle Filter Height, Max. Motor Vehicle Filter Width, Min. Motor Vehicle Filter Height, Min. Motor Vehicle Filter Width

Illegal Fueling Detection: Enable Work Clothes

Campus Vehicle Overspeed Detection: Default Speed Limit, Speed Limit Ratio of Speeding, Speed Correction

Campus Vehicle Congestion Detection: Duration, Vehicle Threshold

Vehicle Driving Away Detection: Time Threshold

Vehicle Congestion Detection: Time Threshold

Litter Detection: Traffic Event Alarm Interval

Road Snow Accumulation Detection: Traffic Event Alarm Interval

Traffic Accident Detection: Time Threshold

Fog Detection: Traffic Event Alarm Interval

Pedestrian Intrusion Detection: Traffic Event Alarm Interval

Abnormal Parking Detection: Time Threshold

Ship Detection: Ship Type

Restricted Area Swimming: Max. Head & Shoulder Filter Height, Max. Head & Shoulder Filter Width, Min. Head & Shoulder Filter Height, Min. Head & Shoulder Filter Width

Water Level Detection: Highest Water Level, Lowest Water Level

Abnormal Water Color Detection: Alarm Color

Oversized Coal Detection: Proportion of the belt width taken up by coal blocks (in ten-thousandths percentage)

Coal Quantity Detection: alarm coal quantity

Unwashed vehicle detection: Note: When EIA device configures/updates smart configurations or reports an alarm, the alarm type passed is UnwashedVehicleDetection (Unwashed Vehicle Detection). However, when retrieving data, the alarm type passed is VehicleBypassingToAvoidWashing (Vehicle Bypassing to Avoid Washing), and InsufficientVehicleWashingWaterVolume (Insufficient Vehicle Washing Water Volume).

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Algorithm Type	Algorithm Name	Area Type	Sensitivity	Alarm Interval	Direction Line	Trigger Time	Person Number Limit	Fluctuating Number of People	Background Refresh Interval	Alarm Reset
ExposedGarbageDetection	Exposed Trash Detection	polygon	√	√		√				

Algorithm Type	Algorithm Name	Area Type	Sensitivity	Alarm Interval	Direction Line	Trigger Time	Person Number Limit	Fluctuating Number of People	Background Refresh Interval	Alarm Reset
OutsideTheStoreBusinessDetection	Out-of-store Object Detection	polygon	√	√		√				
PackedGarbageDetection	Bagged Trash Detection	polygon	√	√		√				
NonMotorAbnormalParkingDetection	Illegally Parked Non-Motor Vehicle Detection	polygon	√	√		√				√
SharedBicyclesRandomParkingDetection	Illegally Parked Shared Bicycle Detection	polygon	√	√		√				
FloatingObjOnRiverDetection	Floating Debris in River Detection	polygon	√	√						
AbnormalParkingDetection	Illegally Parked Motor Vehicle Detection	polygon	√	√		√				√
WasteTruckDetection	Dump Trunk Detection	polygon	√	√						√
TrashOverflowingDetection	Trash Bin Overflow Detection	polygon	√	√		√				
UncoveredTrashCanDetection	Trash Bin Open Detection	polygon	√	√		√				
SetFileOnTheRoadDetection	Burning at	polygon	√	√						√

Algorithm Type	Algorithm Name	Area Type	Sensitivity	Alarm Interval	Direction Line	Trigger Time	Person Number Limit	Fluctuating Number of People	Background Refresh Interval	Alarm Reset
	Intersection Detection									
DirtyRoadDetection	Unclean Road Surface Detection	polygon	✓	✓		✓				
SurfaceWaterDetection	Accumulated Water on Road Detection	polygon	✓	✓		✓				
DogDetection	Dog Detection	polygon	✓	✓						
PeopleGathering	People Gathering	polygon					✓			
IllegalUseOfAnUmbrellaDetection	Illegal Umbrella Shelter Detection	polygon	✓	✓		✓				
IllegalBillboardsDetection	Illegal Billboard Detection	polygon	✓	✓						
DisorderStackingDetection	Improper Stacking of Materials Detection	polygon	✓	✓		✓				
DryingOnTheStreetDetection	Hanging Clothes Along Street Detection	polygon	✓	✓						
StreetVendorDetection	Peddler Detection	polygon	✓	✓		✓				

Algorithm Type	Algorithm Name	Area Type	Sensitivity	Alarm Interval	Direction Line	Trigger Time	Person Number Limit	Fluctuating Number of People	Background Refresh Interval	Alarm Reset
RoadsideStallBusinessDetection	Roadside Booth Detection	polygon	√	√		√				